第六章　重积分及其应用

基础题

一、选择题

(1)B.

解 D 如图 6-1 所示,在 D 上有

$$[\ln(x+y)]^9 \leqslant [\sin(x+y)]^9 \leqslant (x+y)^9,$$

故选 B.

(2)C.

解 依题意,D 如图 6-2 所示,

$$I = \iint_D (xy + \cos x \sin y)\mathrm{d}x\mathrm{d}y = \iint_D xy\mathrm{d}x\mathrm{d}y + \iint_D \cos x \sin y\mathrm{d}x\mathrm{d}y = I_1 + I_2.$$

对于 I_1,$D_1 \cup D_2$ 关于 y 轴对称,xy 关于 x 为奇函数,故 $\iint\limits_{D_1 \cup D_2} xy\mathrm{d}x\mathrm{d}y = 0$. 同理,$\iint\limits_{D_3 \cup D_4} xy\mathrm{d}x\mathrm{d}y = 0$.

于是 $I_1 = 0$.

对于 I_2,$D_3 \cup D_4$ 关于 x 轴对称,$\cos x \sin y$ 关于 y 是奇函数,故 $\iint\limits_{D_3 \cup D_4} \cos x \sin y\mathrm{d}x\mathrm{d}y = 0$,

$D_1 \cup D_2$ 关于 y 轴对称,$\cos x \sin y$ 关于 x 是偶函数,故

$$\iint_{D_1 \cup D_2} \cos x \sin y\mathrm{d}x\mathrm{d}y = 2\iint_{D_1} \cos x \sin y\mathrm{d}x\mathrm{d}y,$$

故选 C.

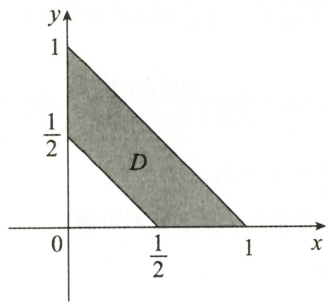

图 6-1

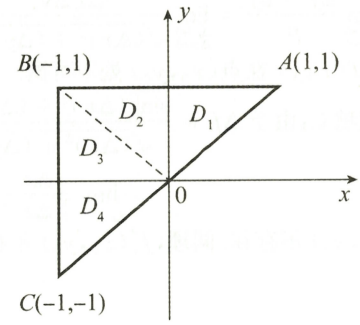

图 6-2

(3)C.

解 原积分 I 的积分区域如图 6-3 所示,

$$D_1 = \begin{cases} 0 \leqslant x \leqslant 2, \\ 0 \leqslant y \leqslant \dfrac{x^2}{2}, \end{cases}$$

$$D_2 = \begin{cases} 2 \leqslant x \leqslant 2\sqrt{2}, \\ 0 \leqslant y \leqslant \sqrt{8-x^2}, \end{cases}$$

故 $I = \int_0^2 \mathrm{d}y \int_{\sqrt{2y}}^{\sqrt{8-y^2}} f(x,y)\mathrm{d}x$. 选项 C 正确.

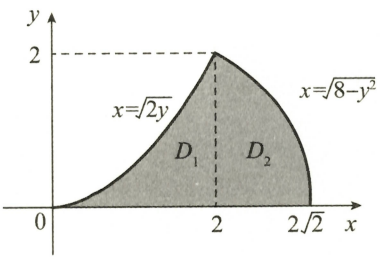

图 6-3

(4)C.

解 $x^2+y^2\leqslant x$ 化为极坐标方程为 $0\leqslant r\leqslant\cos\theta$，$-\frac{\pi}{2}\leqslant\theta\leqslant\frac{\pi}{2}$，故选 C.

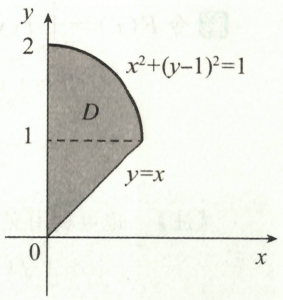

图 6-4

(5)C.

解 由 $r=2\sin\theta$，得 $r^2=2r\sin\theta$，即

$$x^2+y^2=2y,$$

积分区域 D 如图 6-4 所示，故选 C.

(6)A.

解 积分区域 V 关于 yOz 面和 xOz 面对称，$z=zx^0y^0$ 关于 x 与 y 为偶函数，故选 A.

二、填空题

(1)$\frac{3}{2}$.

解 积分区域如图 6-5 所示，交换积分顺序.

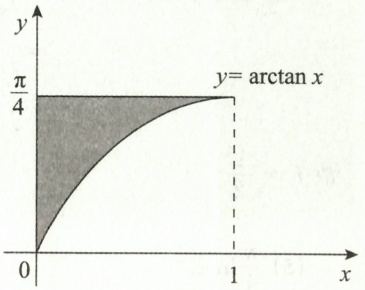

图 6-5

$$\begin{aligned}I&=\int_0^{\frac{\pi}{4}}\csc 2y\,dy\int_0^{\tan y}\frac{1}{3}x^{-\frac{2}{3}}dx\\&=\int_0^{\frac{\pi}{4}}\csc 2y\cdot x^{\frac{1}{3}}\Big|_0^{\tan y}dy\\&=\int_0^{\frac{\pi}{4}}\frac{\tan^{\frac{1}{3}}y}{\sin 2y}dy\\&=\int_0^{\frac{\pi}{4}}\frac{\tan^{\frac{1}{3}}y}{2\tan y\cdot\cos^2 y}dy\\&=\frac{1}{2}\int_0^{\frac{\pi}{4}}\tan^{-\frac{2}{3}}y\,d(\tan y)=\frac{3}{2}\tan^{\frac{1}{3}}y\Big|_0^{\frac{\pi}{4}}=\frac{3}{2}.\end{aligned}$$

(2)e^{-1}.

解 依题设，有 $\int_0^t dx\int_x^t e^{-(x-y)^2}dy$，知积分区域如图 6-6 所示，交换积分顺序，得

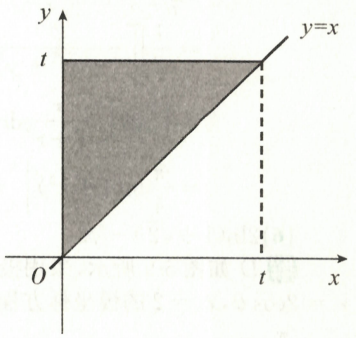

图 6-6

$$f(t)=\int_0^t dy\int_0^y e^{-(x-y)^2}dx,$$

则

$$f'(t)=\int_0^t e^{-(x-t)^2}dx\xlongequal{x-t=u}\int_{-t}^0 e^{-u^2}du=-\int_0^{-t}e^{-u^2}du,$$

$$f''(t)=-e^{-(-t)^2}\cdot(-1)=e^{-t^2},$$

故 $f''(1)=e^{-1}$.

【注】 $f(t)=\int_0^t dy\int_0^y e^{-(x-y)^2}dx=\int_0^t\left[\int_0^y e^{-(x-y)^2}dx\right]dy$，若记 $g(y)=\int_0^y e^{-(x-y)^2}dx$，则 $f(t)=\int_0^t g(y)dy$，故 $f'(t)=g(t)=\int_0^t e^{-(x-t)^2}dx$.

(3)$\frac{a+b}{2}\pi$.

解 如图 6-7 所示，D 关于直线 $y=x$ 对称，故

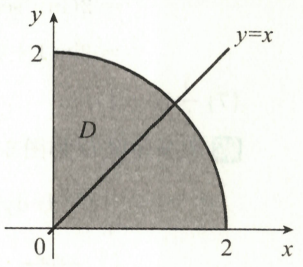

图 6-7

$$\begin{aligned}I&=\frac{1}{2}\iint_D\left[\frac{a\sqrt{f(x)}+b\sqrt{f(y)}}{\sqrt{f(x)}+\sqrt{f(y)}}+\frac{a\sqrt{f(y)}+b\sqrt{f(x)}}{\sqrt{f(y)}+\sqrt{f(x)}}\right]dx\,dy\\&=\frac{a+b}{2}\iint_D dx\,dy=\frac{a+b}{2}\cdot\frac{1}{4}\pi\cdot 2^2=\frac{a+b}{2}\pi.\end{aligned}$$

(4)$\frac{A^2}{2}$.

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解 令 $F(x)=\int_x^1 f(y)\mathrm{d}y$,则 $F'(x)=-f(x)$,故

$$I=\int_0^1\mathrm{d}x\int_x^1 f(x)f(y)\mathrm{d}y=\int_0^1 f(x)\mathrm{d}x\int_x^1 f(y)\mathrm{d}y$$
$$=-\int_0^1 F(x)\mathrm{d}[F(x)]=-\frac{1}{2}F^2(x)\Big|_0^1=\frac{A^2}{2}.$$

【注】 也可利用交换积分顺序及积分与积分变量无关求解.

$$\int_0^1\mathrm{d}x\int_x^1 f(x)f(y)\mathrm{d}y=\int_0^1\mathrm{d}y\int_0^y f(x)f(y)\mathrm{d}x=\int_0^1\mathrm{d}x\int_0^x f(x)f(y)\mathrm{d}y,$$

故

$$2\int_0^1\mathrm{d}x\int_x^1 f(x)f(y)\mathrm{d}y=\int_0^1\mathrm{d}x\int_0^x f(x)f(y)\mathrm{d}y+\int_0^1\mathrm{d}x\int_x^1 f(x)f(y)\mathrm{d}y$$
$$=\int_0^1\mathrm{d}x\left\{f(x)\left[\int_0^x f(y)\mathrm{d}y+\int_x^1 f(y)\mathrm{d}y\right]\right\}$$
$$=\int_0^1\mathrm{d}x\int_0^1 f(x)f(y)\mathrm{d}y=\left[\int_0^1 f(x)\mathrm{d}x\right]\left[\int_0^1 f(y)\mathrm{d}y\right]=A^2,$$

故 $I=\dfrac{A^2}{2}$.

(5) $\dfrac{\pi}{4}\ln 2$.

解 如图 6-8 所示,由于 D 关于直线 $y=x$ 对称,所以

$$I=\iint_D\frac{1+x-y}{1+x^2+y^2}\mathrm{d}x\,\mathrm{d}y=\frac{1}{2}\iint_D\left(\frac{1+x-y}{1+x^2+y^2}+\frac{1+y-x}{1+y^2+x^2}\right)\mathrm{d}x\,\mathrm{d}y$$
$$=\frac{1}{2}\iint_D\frac{2}{1+x^2+y^2}\mathrm{d}x\,\mathrm{d}y$$
$$=\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^1\frac{r}{1+r^2}\mathrm{d}r=\frac{\pi}{2}\cdot\frac{1}{2}\int_0^1\frac{\mathrm{d}(1+r^2)}{1+r^2}$$
$$=\frac{\pi}{4}\ln(1+r^2)\Big|_0^1=\frac{\pi}{4}\ln 2.$$

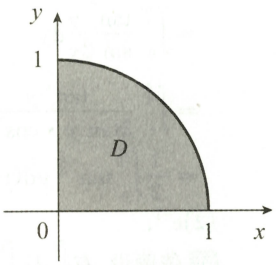

图 6-8

(6) $2\ln(1+\sqrt{2})-\sqrt{2}$.

解 D 如图 6-9 所示,采用极坐标. $x^2+y^2=2x$ 的极坐标方程为 $r=2\cos\theta$,$x=2$ 的极坐标方程为 $r=2\sec\theta$,$y=x$ 的极坐标方程为 $\theta=\dfrac{\pi}{4}$,故

$$I=\iint_D\frac{\mathrm{d}x\,\mathrm{d}y}{\sqrt{x^2+y^2}}=\int_0^{\frac{\pi}{4}}\mathrm{d}\theta\int_{2\cos\theta}^{2\sec\theta}\frac{1}{r}\cdot r\mathrm{d}r$$
$$=2\int_0^{\frac{\pi}{4}}(\sec\theta-\cos\theta)\mathrm{d}\theta$$
$$=2(\ln|\sec\theta+\tan\theta|-\sin\theta)\Big|_0^{\frac{\pi}{4}}$$
$$=2\ln(1+\sqrt{2})-\sqrt{2}.$$

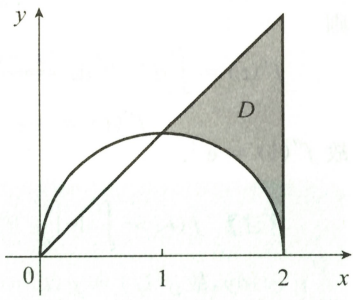

图 6-9

(7) $\dfrac{1}{2}(\mathrm{e}-1)$.

解 积分区域 D 如图 6-10 所示,采用极坐标.

$$I=\iint_D\mathrm{e}^{\frac{x}{x+y}}\mathrm{d}x\,\mathrm{d}y=\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{\frac{1}{\cos\theta+\sin\theta}}\mathrm{e}^{\frac{\cos\theta}{\cos\theta+\sin\theta}}r\mathrm{d}r$$
$$=\int_0^{\frac{\pi}{2}}\mathrm{e}^{\frac{\cos\theta}{\cos\theta+\sin\theta}}\cdot\frac{1}{2}r^2\Big|_0^{\frac{1}{\cos\theta+\sin\theta}}\mathrm{d}\theta$$

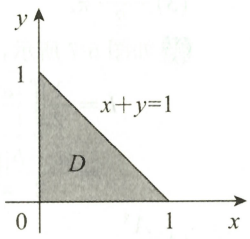

图 6-10

$$=\frac{1}{2}\int_0^{\frac{\pi}{2}} e^{\frac{\cos\theta}{\cos\theta+\sin\theta}}\cdot\frac{1}{(\cos\theta+\sin\theta)^2}d\theta$$

$$=-\frac{1}{2}\int_0^{\frac{\pi}{2}} e^{\frac{\cos\theta}{\cos\theta+\sin\theta}}d\left(\frac{\cos\theta}{\cos\theta+\sin\theta}\right)$$

$$=-\frac{1}{2}e^{\frac{\cos\theta}{\cos\theta+\sin\theta}}\Big|_0^{\frac{\pi}{2}}=\frac{1}{2}(e-1).$$

(8) $\dfrac{13\pi}{144}$.

解 D 关于直线 $y=x$ 对称，由轮换对称性，有

$$I=\iint_D\left(\frac{x^2}{4}+\frac{y^2}{9}\right)dx\,dy=\frac{1}{2}\iint_D\left(\frac{x^2}{4}+\frac{y^2}{9}+\frac{y^2}{4}+\frac{x^2}{9}\right)dx\,dy$$

$$=\frac{1}{2}\left(\frac{1}{4}+\frac{1}{9}\right)\iint_D(x^2+y^2)dx\,dy$$

$$=\frac{1}{2}\left(\frac{1}{4}+\frac{1}{9}\right)\int_0^{2\pi}d\theta\int_0^1 r^3dr=\frac{13\pi}{144}.$$

(9) $4-\dfrac{\pi}{2}$.

解 如图 6-11 所示，设

$$D_{大}=\{(x,y)\,|\,-2\leqslant x\leqslant 0,\ 0\leqslant y\leqslant 2\},$$

$$D_{小}=\{(x,y)\,|\,-\sqrt{2y-y^2}\leqslant x\leqslant 0\},$$

则

$$I=\iint_D y\,dx\,dy=\iint_{D_{大}}y\,dx\,dy-\iint_{D_{小}}y\,dx\,dy$$

$$=\int_{-2}^0 dx\int_0^2 y\,dy-\int_{\frac{\pi}{2}}^{\pi}d\theta\int_0^{2\sin\theta}r^2\sin\theta\,dr$$

$$=4-\frac{8}{3}\int_{\frac{\pi}{2}}^{\pi}\sin^4\theta\,d\theta=4-\frac{8}{3}\int_0^{\frac{\pi}{2}}\cos^4 t\,dt$$

$$=4-\frac{8}{3}\times\frac{3}{4}\times\frac{1}{2}\times\frac{\pi}{2}=4-\frac{\pi}{2}.$$

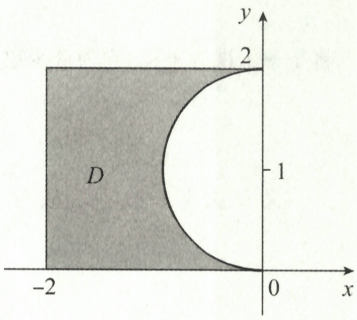

图 6-11

【注】 由于 D 关于 $y=1$ 对称，故

$$I=\iint_D y\,dx\,dy=\iint_D[(y-1)+1]dx\,dy=\iint_D(y-1)dx\,dy+\iint_D dx\,dy$$

$$=0+2^2-\frac{1}{2}\pi\times 1^2=4-\frac{\pi}{2},$$

或利用形心纵坐标为 $\bar{y}=1$，有 $I=\iint_D y\,dx\,dy=\bar{y}\cdot\left(4-\dfrac{\pi}{2}\right)=4-\dfrac{\pi}{2}$.

(10) 2π.

解 D 如图 6-12 所示，利用极坐标，

$$I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}d\theta\int_0^{2\cos\theta}(2r\cos\theta+3r\sin\theta)r\,dr$$

$$=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{16}{3}\cos^4\theta+8\cos^3\theta\cdot\sin\theta\right)d\theta$$

$$=\frac{32}{3}\int_0^{\frac{\pi}{2}}\cos^4\theta\,d\theta+0$$

$$=\frac{32}{3}\times\frac{3}{4}\times\frac{1}{2}\times\frac{\pi}{2}=2\pi.$$

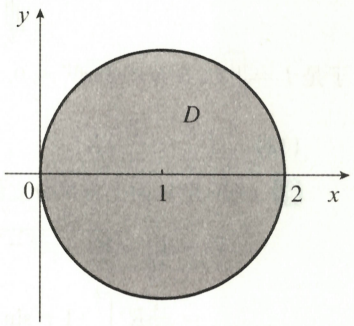

图 6-12

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【注】 ① 考虑到被积函数关于 x,y 都是一次函数,可利用形心坐标 $(\overline{x},\overline{y})=(1,0)$.

由

$$\overline{x}=\frac{\iint\limits_D x\rho\,\mathrm{d}x\,\mathrm{d}y}{\iint\limits_D \rho\,\mathrm{d}x\,\mathrm{d}y}=\frac{\iint\limits_D x\,\mathrm{d}x\,\mathrm{d}y}{\iint\limits_D \mathrm{d}x\,\mathrm{d}y},$$

有 $\iint\limits_D x\,\mathrm{d}x\,\mathrm{d}y=\overline{x}\cdot\iint\limits_D \mathrm{d}x\,\mathrm{d}y$,其中 $\iint\limits_D \mathrm{d}x\,\mathrm{d}y$ 表示 D 的面积.

同理,$\iint\limits_D y\,\mathrm{d}x\,\mathrm{d}y=\overline{y}\cdot\iint\limits_D \mathrm{d}x\,\mathrm{d}y$,所以

$$I=\iint\limits_D (2x+3y)\,\mathrm{d}x\,\mathrm{d}y=(2\overline{x}+3\overline{y})\pi\cdot 1^2=2\pi.$$

② 考虑到 D 关于直线 $x=1$ 对称,则

$$I=\iint\limits_D (2x+3y)\,\mathrm{d}x\,\mathrm{d}y=\iint\limits_D [2(x-1)+2+3y]\,\mathrm{d}x\,\mathrm{d}y,$$

将 $x-1$ 视为整体,它为奇函数,故 $\iint\limits_D 2(x-1)\,\mathrm{d}x\,\mathrm{d}y=0$,所以

$$I=2\iint\limits_D \mathrm{d}x\,\mathrm{d}y+3\iint\limits_D y\,\mathrm{d}x\,\mathrm{d}y.$$

又 D 关于 x 轴对称,故 $3\iint\limits_D y\,\mathrm{d}x\,\mathrm{d}y=0$,于是

$$I=2\iint\limits_D \mathrm{d}x\,\mathrm{d}y=2\times\pi\times 1^2=2\pi.$$

(11) 0.

解 V 是以 $(-1,1,0)$ 为球心、$\sqrt{2}$ 为半径的球体,若视其密度为常数,则质心(形心)坐标为 $(-1,1,0)$,则

$$\overline{x}=\frac{\iiint\limits_V x\,\mathrm{d}V}{\iiint\limits_V \mathrm{d}V},\quad \overline{y}=\frac{\iiint\limits_V y\,\mathrm{d}V}{\iiint\limits_V \mathrm{d}V},\quad \overline{z}=\frac{\iiint\limits_V z\,\mathrm{d}V}{\iiint\limits_V \mathrm{d}V}.$$

而 $\iiint\limits_V \mathrm{d}V$ 为球的体积 $\frac{8}{3}\sqrt{2}\pi$,故

$$\iiint\limits_V x\,\mathrm{d}V=\frac{8}{3}\sqrt{2}\pi\times(-1)=-\frac{8}{3}\sqrt{2}\pi,$$

$$\iiint\limits_V y\,\mathrm{d}V=\frac{8}{3}\sqrt{2}\pi\times 1=\frac{8}{3}\sqrt{2}\pi,$$

$$\iiint\limits_V z\,\mathrm{d}V=\frac{8}{3}\sqrt{2}\pi\times 0=0,$$

于是 $I=\iiint\limits_V (x+y+z)\,\mathrm{d}V=0$.

(12) $\frac{4}{3}\left(\frac{\pi}{2}-\frac{2}{3}\right)R^3$.

解 xOy 坐标面上方部分立体如图 6-13 所示,则

$$V=4\iint\limits_D \sqrt{R^2-x^2-y^2}\,\mathrm{d}x\,\mathrm{d}y=4\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{R\cos\theta}\sqrt{R^2-r^2}\cdot r\,\mathrm{d}r$$

$$=\frac{4}{3}R^3\int_0^{\frac{\pi}{2}}(1-\sin^3\theta)\,\mathrm{d}\theta=\frac{4}{3}\left(\frac{\pi}{2}-\frac{2}{3}\right)R^3.$$

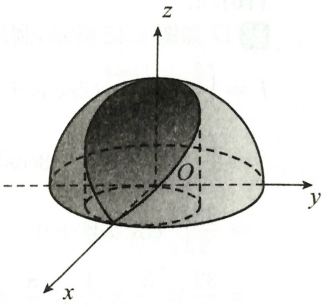


图 6-13

(13) $\left(\dfrac{15\pi-32}{30\pi-48},0\right)$.

解 如图 6-14 所示，由对称性，知 $\bar{y}=0$. 又由于

$$\bar{x}=\frac{\iint\limits_D x\,\mathrm{d}x\,\mathrm{d}y}{\iint\limits_D \mathrm{d}x\,\mathrm{d}y}=\frac{2\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^1 r\cos\theta\cdot r\,\mathrm{d}r+2\int_{\frac{\pi}{2}}^{\pi}\mathrm{d}\theta\int_0^{1+\cos\theta} r\cos\theta\cdot r\,\mathrm{d}r}{\frac{1}{2}\pi\times1^2+2\int_{\frac{\pi}{2}}^{\pi}\mathrm{d}\theta\int_0^{1+\cos\theta} r\,\mathrm{d}r}$$

$$=\frac{2\int_0^{\frac{\pi}{2}}\cos\theta\,\mathrm{d}\theta\int_0^1 r^2\,\mathrm{d}r+2\int_{\frac{\pi}{2}}^{\pi}\cos\theta\,\mathrm{d}\theta\int_0^{1+\cos\theta} r^2\,\mathrm{d}r}{\frac{1}{2}\pi+2\cdot\frac{1}{2}\int_{\frac{\pi}{2}}^{\pi}\left(r^2\Big|_0^{1+\cos\theta}\right)\mathrm{d}\theta}$$

$$=\frac{\frac{2}{3}\left(\frac{15\pi}{16}-2\right)}{\frac{5}{4}\pi-2}=\frac{15\pi-32}{30\pi-48},$$

故形心坐标为 $\left(\dfrac{15\pi-32}{30\pi-48},0\right)$.

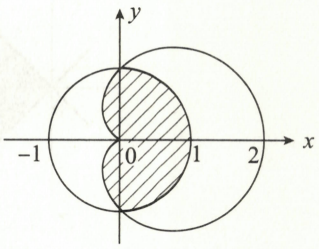


图 6-14

(14) $\dfrac{1}{44}$，$\dfrac{1}{36}$.

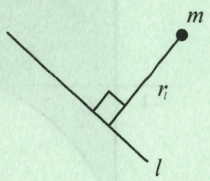
解 对 x 轴与 y 轴的转动惯量分别为 $I_x=\iint\limits_D y^2\,\mathrm{d}x\,\mathrm{d}y$ 与 $I_y=\iint\limits_D x^2\,\mathrm{d}x\,\mathrm{d}y$，故

$$I_x=\iint\limits_D y^2\,\mathrm{d}x\,\mathrm{d}y=\int_0^1\mathrm{d}x\int_{x^{\frac{3}{2}}}^{x} y^2\,\mathrm{d}y=\int_0^1\frac{1}{3}y^3\Big|_{x^{\frac{3}{2}}}^{x}\mathrm{d}x=\frac{1}{44},$$

$$I_y=\iint\limits_D x^2\,\mathrm{d}x\,\mathrm{d}y=\int_0^1\mathrm{d}x\int_{x^{\frac{3}{2}}}^{x} x^2\,\mathrm{d}y=\frac{1}{36}.$$

【注】 质点(质量为 m)对轴 l 的转动惯量为 $I_l=m\cdot r_l^2$，其中 r_l 表示质点到轴 l 的距离.

平面区域 D，对 x 轴的转动惯量可以采用微元法：任取一个小区域 $\mathrm{d}\sigma$($\mathrm{d}\sigma$ 表示面积)，如图 6-15 所示，看成质量集中为一点(x,y)，则微元 $\mathrm{d}I_x=\rho\cdot y^2\mathrm{d}\sigma$($\rho$ 表示 D 的面密度)，故 $I_x=\iint\limits_D y^2\cdot\rho\,\mathrm{d}\sigma$.



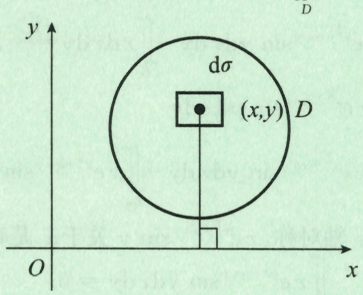


图 6-15

三、解答题

(1) **解** (Ⅰ)D 如图 6-16 所示，先对 y 积分较简便.

$$I=\iint\limits_D xy(x-y)\,\mathrm{d}x\,\mathrm{d}y=\int_0^1\mathrm{d}x\int_{-x}^{x}xy(x-y)\,\mathrm{d}y$$

$$=\int_0^1\left(\frac{x^2y^2}{2}-\frac{xy^3}{3}\right)\Big|_{-x}^{x}\mathrm{d}x=-\frac{2}{3}\int_0^1 x^4\,\mathrm{d}x=-\frac{2}{15}.$$

(Ⅱ)D 如图 6-17 所示，若先对 y 积分，则 $\int\dfrac{\sin y}{y}\mathrm{d}y$ 不能表示为初等函数，故只能先对 x 积分.

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$$I=\iint_D \frac{\sin y}{y}\mathrm{d}x\,\mathrm{d}y=\int_0^1 \mathrm{d}y\int_{y^2}^{y}\frac{\sin y}{y}\mathrm{d}x=\int_0^1 \frac{\sin y}{y}(y-y^2)\mathrm{d}y=\int_0^1 (1-y)\sin y\mathrm{d}y$$
$$=[-(1-y)\cos y-\sin y]\Big|_0^1=1-\sin 1.$$

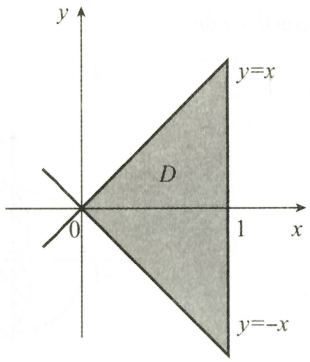

图 6-16

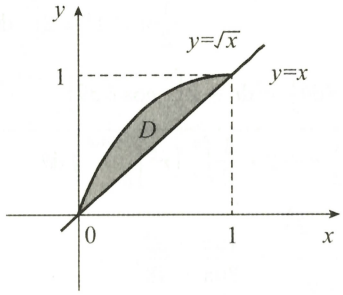

图 6-17

(Ⅲ)D 如图 6-18 所示,先对 x 积分,则

$$I=\iint_D \frac{xy}{\sqrt{1+y^3}}\mathrm{d}x\,\mathrm{d}y$$
$$=\int_0^1 \frac{y}{\sqrt{1+y^3}}\mathrm{d}y\int_0^{\sqrt{y}} x\,\mathrm{d}x$$
$$=\frac{1}{2}\int_0^1 \frac{y^2}{\sqrt{1+y^3}}\mathrm{d}y$$
$$=\frac{1}{2}\cdot\frac{2}{3}(1+y^3)^{\frac{1}{2}}\Big|_0^1$$
$$=\frac{1}{3}(\sqrt{2}-1).$$

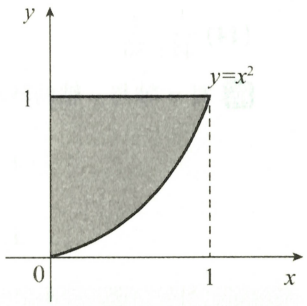

图 6-18

(Ⅳ)D 如图 6-19 所示,作辅助线 $y=-\arcsin x\ (-1\leqslant x\leqslant 0)$,将 D 划分为 D_1 与 D_2,则

$$I=\iint_D x(\mathrm{e}^{x^2+\cos y}\sin y-1)\mathrm{d}x\,\mathrm{d}y$$
$$=\iint_D x\,\mathrm{e}^{x^2+\cos y}\sin y\mathrm{d}x\,\mathrm{d}y-\iint_D x\,\mathrm{d}x\,\mathrm{d}y\overset{\text{记}}{=\!=}I_1-I_2,$$
$$I_1=\iint_D x\,\mathrm{e}^{x^2+\cos y}\sin y\mathrm{d}x\,\mathrm{d}y$$
$$=\iint_{D_1} x\,\mathrm{e}^{x^2+\cos y}\sin y\mathrm{d}x\,\mathrm{d}y+\iint_{D_2} x\,\mathrm{e}^{x^2+\cos y}\sin y\mathrm{d}x\,\mathrm{d}y.$$

由 D_1 关于 y 轴对称,$x\,\mathrm{e}^{x^2+\cos y}\sin y$ 关于 x 是奇函数,故

$$\iint_{D_1} x\,\mathrm{e}^{x^2+\cos y}\sin y\mathrm{d}x\,\mathrm{d}y=0.$$

同理,$\iint_{D_2} x\,\mathrm{e}^{x^2+\cos y}\sin y\mathrm{d}x\,\mathrm{d}y=0$,故 $I_1=0$.

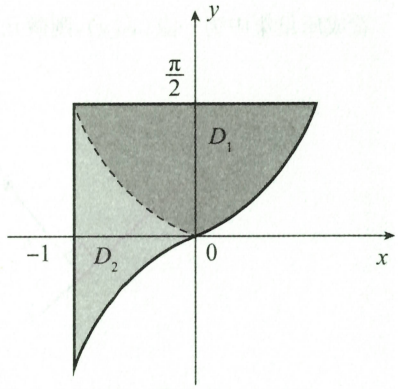

图 6-19

$$I_2=\iint_D x\,\mathrm{d}x\,\mathrm{d}y=\iint_{D_1} x\,\mathrm{d}x\,\mathrm{d}y+\iint_{D_2} x\,\mathrm{d}x\,\mathrm{d}y,$$

根据对称性,$\iint_{D_1} x\,\mathrm{d}x\,\mathrm{d}y=0$. 又

$$\iint_{D_2} x\,\mathrm{d}x\,\mathrm{d}y=2\int_0^{\frac{\pi}{2}}\mathrm{d}y\int_{-1}^{-\sin y} x\,\mathrm{d}x=-\frac{\pi}{4},$$

故 $I=0-\left(-\dfrac{\pi}{4}\right)=\dfrac{\pi}{4}$.

(2) 解 由已知，D 如图 6-20 所示，D 关于直线 $y=x$ 对称，则

$$I=\iint_D \frac{x\mathrm{e}^{(x+y)^2}}{x+y}\mathrm{d}x\mathrm{d}y=\frac{1}{2}\iint_D\left[\frac{x\mathrm{e}^{(x+y)^2}}{x+y}+\frac{y\mathrm{e}^{(y+x)^2}}{y+x}\right]\mathrm{d}x\mathrm{d}y$$

$$=\frac{1}{2}\iint_D \mathrm{e}^{(x+y)^2}\mathrm{d}x\mathrm{d}y$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_{\frac{1}{\cos\theta+\sin\theta}}^{\frac{2}{\cos\theta+\sin\theta}}\mathrm{e}^{r^2(\cos\theta+\sin\theta)^2}\cdot r\mathrm{d}r$$

$$=\frac{1}{4}\int_0^{\frac{\pi}{2}}\left[\frac{1}{(\cos\theta+\sin\theta)^2}\mathrm{e}^{r^2(\cos\theta+\sin\theta)^2}\Big|_{\frac{1}{\cos\theta+\sin\theta}}^{\frac{2}{\cos\theta+\sin\theta}}\right]\mathrm{d}\theta$$

$$=\frac{1}{4}(\mathrm{e}^4-\mathrm{e})\int_0^{\frac{\pi}{2}}\frac{\mathrm{d}\theta}{(\cos\theta+\sin\theta)^2}$$

$$=\frac{1}{4}(\mathrm{e}^4-\mathrm{e})\int_0^{\frac{\pi}{2}}\frac{\mathrm{d}(\tan\theta)}{(1+\tan\theta)^2}=\frac{1}{4}(\mathrm{e}^4-\mathrm{e})\cdot\frac{-1}{1+\tan\theta}\Big|_0^{\frac{\pi}{2}}=\frac{\mathrm{e}^4-\mathrm{e}}{4}.$$

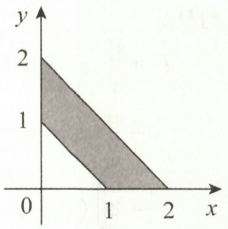

图 6-20

(3) 解 D 如图 6-21 阴影部分所示，D 关于直线 $y=x$ 对称.

$$I=\iint_D(2x^2-y^2)\mathrm{d}x\mathrm{d}y=\iint_D x^2\mathrm{d}x\mathrm{d}y+\iint_D(x^2-y^2)\mathrm{d}x\mathrm{d}y$$

$$\xlongequal{\text{记}} I_1+I_2,$$

$$I_2=\iint_D(x^2-y^2)\mathrm{d}x\mathrm{d}y=\frac{1}{2}\iint_D(x^2-y^2+y^2-x^2)\,\mathrm{d}x\mathrm{d}y=0,$$

$$I_1=\iint_D x^2\mathrm{d}x\mathrm{d}y=\frac{1}{2}\iint_D(x^2+y^2)\mathrm{d}x\mathrm{d}y=\frac{1}{2}\cdot 2\iint_{D_1}(x^2+y^2)\mathrm{d}x\mathrm{d}y,$$

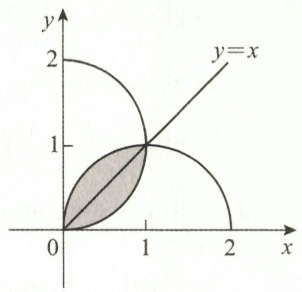

图 6-21

其中 $D_1=\left\{(x,y)\,|\,x^2+(y-1)^2\leqslant 1,0\leqslant y\leqslant x\right\}$.

故

$$I=I_1=\int_0^{\frac{\pi}{4}}\mathrm{d}\theta\int_0^{2\sin\theta}r^2\cdot r\mathrm{d}r=\int_0^{\frac{\pi}{4}}\frac{1}{4}r^4\Big|_0^{2\sin\theta}\mathrm{d}\theta$$

$$=4\int_0^{\frac{\pi}{4}}\sin^4\theta\mathrm{d}\theta=\int_0^{\frac{\pi}{4}}(1-\cos 2\theta)^2\mathrm{d}\theta$$

$$\xlongequal{2\theta=t}\int_0^{\frac{\pi}{2}}(1-\cos t)^2\cdot\frac{1}{2}\mathrm{d}t$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{2}}(1-2\cos t+\cos^2 t)\,\mathrm{d}t$$

$$=\frac{1}{2}\left(\frac{\pi}{2}-2+\frac{1}{2}\cdot\frac{\pi}{2}\right)=\frac{3}{8}\pi-1.$$

(4) 解 用 $x^2+y^2=1$，即 $r=1$ 将 D 划分为 D_1 与 D_2，如图 6-22 所示，则

$$I=\iiint_{D_1}(1-r)r\mathrm{d}r\mathrm{d}\theta+\iiint_{D_2}(r-1)r\mathrm{d}r\mathrm{d}\theta$$

$$=\iiint_{D_1}(1-r)r\mathrm{d}r\mathrm{d}\theta-\iiint_{D_2}(1-r)r\mathrm{d}r\mathrm{d}\theta$$

$$=2\iiint_{D_1}(1-r)r\mathrm{d}r\mathrm{d}\theta-\iiint_{D}(1-r)r\mathrm{d}r\mathrm{d}\theta$$

$$=2\int_0^{\frac{\pi}{4}}\mathrm{d}\theta\int_0^1(1-r)r\mathrm{d}r-\int_0^{\frac{\pi}{4}}\mathrm{d}\theta\int_0^{\sqrt{2}\cos\theta}(1-r)r\mathrm{d}r$$

$$=\frac{\pi}{12}-\int_0^{\frac{\pi}{4}}\left(\cos^2\theta-\frac{2\sqrt{2}}{3}\cos^3\theta\right)\mathrm{d}\theta$$

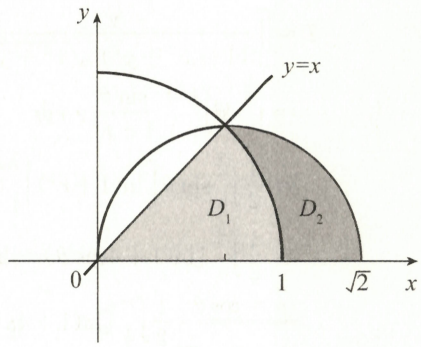

图 6-22

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$$=\frac{\pi}{12}+\frac{5}{9}-\frac{1}{4}-\frac{\pi}{8}=\frac{11}{36}-\frac{\pi}{24}.$$

(5) **解** 积分区域 D 为图 6-23 中的半圆区域，$x+y-2=0$ 将 D 分成 D_1 与 D_2.

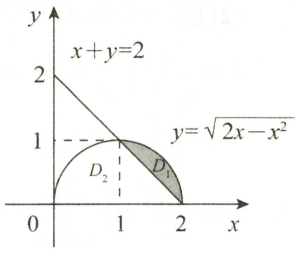

图 6-23

$$I=\iint_D|x+y-2|\,dx\,dy$$

$$=\iint_{D_1}(x+y-2)\,dx\,dy-\iint_{D_2}(x+y-2)\,dx\,dy$$

$$=2\iint_{D_1}(x+y-2)\,dx\,dy-\iint_{D_1+D_2}(x+y-2)\,dx\,dy\overset{记}{=\!=}I_1-I_2,$$

$$I_1=2\iint_{D_1}(x+y-2)\,dx\,dy\quad(\text{先对 }x\text{ 积分后对 }y\text{ 积分})$$

$$=2\int_0^1 dy\int_{2-y}^{1+\sqrt{1-y^2}}(x+y)\,dx-4\iint_{D_1}dx\,dy$$

$$=2\int_0^1\left[\frac{(1+\sqrt{1-y^2})^2-(2-y)^2}{2}+y(1+\sqrt{1-y^2}-2+y)\right]dy-4\left(\frac{1}{4}\pi\times1^2-\frac{1}{2}\right)$$

$$=2\int_0^1(y\sqrt{1-y^2}+\sqrt{1-y^2}+y-1)\,dy-(\pi-2)$$

$$=2\left(\frac{1}{3}+\frac{\pi}{4}-\frac{1}{2}\right)-(\pi-2)=\frac{5}{3}-\frac{\pi}{2},$$

$$I_2=\iint_D(x+y-2)\,dx\,dy=\iint_D(x+y)\,dx\,dy-\iint_D 2\,dx\,dy$$

$$=\int_0^{\frac{\pi}{2}}d\theta\int_0^{2\cos\theta}r^2(\cos\theta+\sin\theta)\,dr-2\times\frac{\pi}{2}\times1^2$$

$$=\frac{1}{3}\int_0^{\frac{\pi}{2}}\left[(\cos\theta+\sin\theta)r^3\Big|_0^{2\cos\theta}\right]d\theta-\pi$$

$$=\frac{8}{3}\int_0^{\frac{\pi}{2}}\cos^4\theta\,d\theta+\frac{8}{3}\int_0^{\frac{\pi}{2}}\cos^3\theta\sin\theta\,d\theta-\pi$$

$$=\frac{8}{3}\times\frac{3}{4}\times\frac{1}{2}\times\frac{\pi}{2}-\frac{8}{3}\times\frac{1}{4}\cos^4\theta\Big|_0^{\frac{\pi}{2}}-\pi$$

$$=\frac{\pi}{2}+\frac{2}{3}-\pi=\frac{2}{3}-\frac{\pi}{2},$$

故 $I=I_1-I_2=\frac{5}{3}-\frac{\pi}{2}-\left(\frac{2}{3}-\frac{\pi}{2}\right)=1.$

(6) **解** D 如图 6-24 所示，由 $\begin{cases}x^2+y^2=1,\\x^2+y^2=2x,\end{cases}$ 解得交点 $A\left(\frac{1}{2},\frac{\sqrt{3}}{2}\right)$.

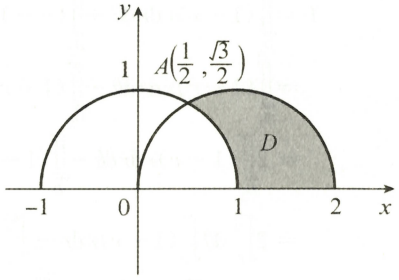

图 6-24

$$I=\iint_D\frac{y}{(1+x^2+y^2)\sqrt{x^2+y^2}}dx\,dy$$

$$=\int_0^{\frac{\pi}{3}}d\theta\int_1^{2\cos\theta}\frac{\sin\theta}{1+r^2}\cdot r\,dr$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{3}}\sin\theta\left[\ln(1+r^2)\Big|_1^{2\cos\theta}\right]d\theta$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{3}}[\ln(1+4\cos^2\theta)-\ln 2]\sin\theta\,d\theta$$

$$\overset{u=\cos\theta}{=\!=}\frac{1}{2}\int_{\frac{1}{2}}^1[\ln(1+4u^2)-\ln 2]\,du$$

$$=\frac{1}{2}\left[u\ln(1+4u^2)-2u+\arctan 2u\right]\Big|_{\frac{1}{2}}^{1}-\frac{1}{4}\ln 2$$

$$=\frac{1}{2}\left(\ln\frac{5}{2}-1+\arctan 2-\frac{\pi}{4}\right).$$

(7) 解 如图 6-25 所示，直线 $x+y=i(i=1,2,3,4)$ 将 D 分为 4 个区域 $D_k(k=1,2,3,4)$，则

$$[1+x+y]=k(k=1,2,3,4),$$

故

$$I=\iint_D[1+x+y]\mathrm{d}x\,\mathrm{d}y$$

$$=\iint_{D_1}1\mathrm{d}x\mathrm{d}y+\iint_{D_2}2\mathrm{d}x\mathrm{d}y+\iint_{D_3}3\mathrm{d}x\mathrm{d}y+\iint_{D_4}4\mathrm{d}x\,\mathrm{d}y$$

$$=10.$$

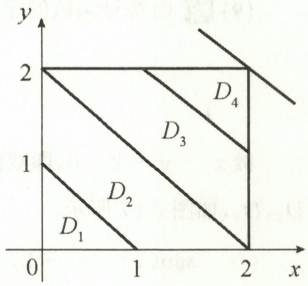

图 6-25

【注】 $\iint_D\mathrm{d}x\,\mathrm{d}y$ 表示 D 的面积.

(8) 解 当 $2x-x^2=(1-y)^2$ 时，有

$$(x-1)^2+(y-1)^2=1.$$

圆周 $(x-1)^2+(y-1)^2=1$ 将 D 分为 D_1 与 D_2 两个区域如图 6-26 所示. 于是

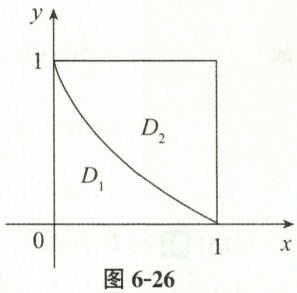

图 6-26

$$\max\{2x-x^2,(1-y)^2\}=\begin{cases}(1-y)^2, & (x,y)\in D_1,\\ 2x-x^2, & (x,y)\in D_2.\end{cases}$$

$$I=\iint_{D_1}(1-y)^2\mathrm{d}x\mathrm{d}y+\iint_{D_2}(2x-x^2)\mathrm{d}x\mathrm{d}y$$

$$=\int_0^1\mathrm{d}y\int_0^{1-\sqrt{2y-y^2}}(1-y)^2\mathrm{d}x+\int_0^1\mathrm{d}x\int_{1-\sqrt{2x-x^2}}^1(2x-x^2)\mathrm{d}y$$

$$=\int_0^1(1-y)^2(1-\sqrt{2y-y^2})\mathrm{d}y+\int_0^1(2x-x^2)[1-(1-\sqrt{2x-x^2})]\mathrm{d}x$$

$$=\int_0^1(1-y)^2\mathrm{d}y-\int_0^1(1-y)^2\sqrt{1-(y-1)^2}\,\mathrm{d}y+\int_0^1(2x-x^2)\sqrt{2x-x^2}\,\mathrm{d}x$$

$$=-\frac{1}{3}(1-y)^3\Big|_0^1-\int_0^1(y-1)^2\sqrt{1-(y-1)^2}\,\mathrm{d}(y-1)+\int_0^1[1-(x-1)^2]\sqrt{1-(x-1)^2}\,\mathrm{d}(x-1)$$

$$=\frac{1}{3}+\int_0^1\sqrt{1-(x-1)^2}\,\mathrm{d}(x-1)-2\int_0^1(x-1)^2\sqrt{1-(x-1)^2}\,\mathrm{d}(x-1),$$

而

$$\int_0^1\sqrt{1-(x-1)^2}\,\mathrm{d}(x-1)\xlongequal{x-1=t}\int_{-1}^0\sqrt{1-t^2}\,\mathrm{d}t\xlongequal{t=-u}\int_1^0\sqrt{1-u^2}(-\mathrm{d}u)$$

$$=\int_0^1\sqrt{1-u^2}\,\mathrm{d}u=\frac{1}{4}\pi\cdot 1^2=\frac{\pi}{4}.$$

$\left(\int_0^1\sqrt{1-u^2}\,\mathrm{d}u\right.$ 表示四分之一圆的面积.$\left.\right)$

$$\int_0^1(x-1)^2\sqrt{1-(x-1)^2}\,\mathrm{d}(x-1)\xlongequal{x-1=\sin t}\int_{-\frac{\pi}{2}}^0\sin^2t\cos t\cos t\,\mathrm{d}t$$

$$\xlongequal{t=-u}\int_{\frac{\pi}{2}}^0\sin^2u\cos^2u(-\mathrm{d}u)=\int_0^{\frac{\pi}{2}}\sin^2u(1-\sin^2u)\mathrm{d}u$$

$$=\int_0^{\frac{\pi}{2}}\sin^2u\,\mathrm{d}u-\int_0^{\frac{\pi}{2}}\sin^4u\,\mathrm{d}u$$

$$=\frac{1}{2}\cdot\frac{\pi}{2}-\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}=\frac{\pi}{4}-\frac{3\pi}{16}.$$

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故 $I=\dfrac{1}{3}+\dfrac{\pi}{4}-2\left(\dfrac{\pi}{4}-\dfrac{3\pi}{16}\right)=\dfrac{1}{3}+\dfrac{\pi}{8}$.

(9) **解** 由符号函数的定义,知

$$\operatorname{sgn}(x^2-y^2+2)=\begin{cases}1, & x^2-y^2+2>0,\\ 0, & x^2-y^2+2=0,\\ -1, & x^2-y^2+2<0,\end{cases}$$

故 $x^2-y^2+2=0$,即双曲线 $\dfrac{y^2}{2}-\dfrac{x^2}{2}=1$ 将 D 划分为三个区域 D_1,D_2,D_3,如图 6-27 所示,

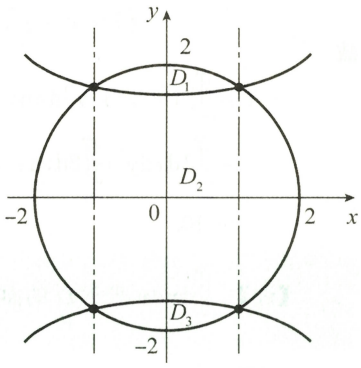

图 6-27

$$\operatorname{sgn}(x^2-y^2+2)=\begin{cases}1, & (x,y)\in D_2,\\ -1, & (x,y)\in D_1\cup D_3,\end{cases}$$

故
$$I=\iint_D\operatorname{sgn}(x^2-y^2+2)\,dx\,dy=\iint_{D_2}1\,dx\,dy-\iint_{D_1}1\,dx\,dy-\iint_{D_3}1\,dx\,dy$$
$$=\iint_{D_2}dx\,dy-2\iint_{D_1}dx\,dy.$$

而
$$\iint_{D_2}dx\,dy=\iint_D dx\,dy-\iint_{D_1\cup D_3}dx\,dy=\iint_D dx\,dy-2\iint_{D_1}dx\,dy,$$

故
$$I=\iint_D dx\,dy-4\iint_{D_1}dx\,dy=\pi\times 2^2-4\int_{-1}^{1}dx\int_{\sqrt{2+x^2}}^{\sqrt{4-x^2}}dy$$
$$=4\pi-4\left[\frac{x}{2}\sqrt{4-x^2}+2\arcsin\frac{x}{2}-\frac{x}{2}\sqrt{2+x^2}-\ln\left(x+\sqrt{2+x^2}\right)\right]\Bigg|_{-1}^{1}$$
$$=4\pi-4\left[\frac{2\pi}{3}-\ln(2+\sqrt{3})\right]=\frac{4\pi}{3}+4\ln(2+\sqrt{3}).$$

(10) **解** 将 D 分成 $D=D_1+D_2+D_3$,如图 6-28 所示,则

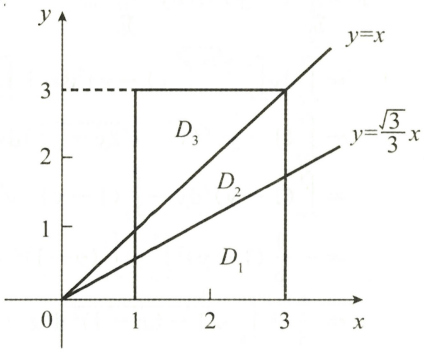

图 6-28

$$I=\iint_D f(x,y)\,dx\,dy=\iint_{D_1}0\,dx\,dy+\iint_{D_2}\frac{dx\,dy}{(x^2+y^2)^2}+\iint_{D_3}0\,dx\,dy$$
$$=\int_{\frac{\pi}{6}}^{\frac{\pi}{4}}d\theta\int_{\sec\theta}^{3\sec\theta}\frac{r}{r^4}dr=\frac{4}{9}\int_{\frac{\pi}{6}}^{\frac{\pi}{4}}\cos^2\theta\,d\theta=\frac{2}{9}\int_{\frac{\pi}{6}}^{\frac{\pi}{4}}(1+\cos 2\theta)\,d\theta$$
$$=\frac{2}{9}\left(\frac{\pi}{12}+\frac{1}{2}-\frac{\sqrt{3}}{4}\right)=\frac{\pi+6-3\sqrt{3}}{54}.$$

(11) **解** 由 $x+y+xy=1$,得 $y=\dfrac{1-x}{1+x}$,即 $y+1=\dfrac{2}{1+x}$,它是以 $x=-1$,$y=-1$ 为渐近线的双曲线,如图 6-29 所示,$x+y=1$ 将 D 分为 D_1 与 D_2,故

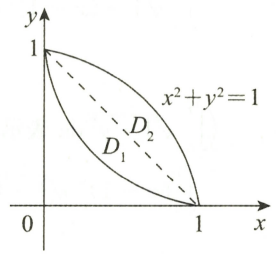

图 6-29

$$I=\iint_{D_1}(x-y+1)\,dx\,dy+\iint_{D_2}\frac{(x+y)^2}{x^2+y^2}dx\,dy$$
$$\xlongequal{\text{记}}I_1+I_2.$$

由于 D_1 关于直线 $y=x$ 对称,所以

$$\iint_{D_1}(x-y)\,dx\,dy=\frac{1}{2}\iint_{D_1}(x-y+y-x)\,dx\,dy=0,$$

$$I_1=0+\iint_{D_1}dx\,dy=\int_0^1 dx\int_{\frac{1-x}{1+x}}^{1-x}dy$$
$$=\int_0^1\left(1-x-\frac{1-x}{1+x}\right)dx=\frac{3}{2}-2\ln 2,$$

$$I_2=\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_{\frac{1}{\cos\theta+\sin\theta}}^{1}\frac{r^2(\cos\theta+\sin\theta)^2}{r^2}\cdot r\,\mathrm{d}r$$

$$=\int_0^{\frac{\pi}{2}}(\cos\theta+\sin\theta)^2\cdot\frac{1}{2}r^2\Big|_{\frac{1}{\cos\theta+\sin\theta}}^{1}\mathrm{d}\theta$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{2}}(\cos\theta+\sin\theta)^2\left[1-\frac{1}{(\cos\theta+\sin\theta)^2}\right]\mathrm{d}\theta$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{2}}(1+\sin 2\theta)\,\mathrm{d}\theta-\frac{1}{2}\int_0^{\frac{\pi}{2}}\mathrm{d}\theta=\frac{1}{2}\int_0^{\frac{\pi}{2}}\sin 2\theta\,\mathrm{d}\theta=\frac{1}{2},$$

故 $I=I_1+I_2=\dfrac{3}{2}-2\ln 2+\dfrac{1}{2}=2-2\ln 2$.

(12) 解 （Ⅰ）双纽线$(x^2+y^2)^2=2(x^2-y^2)$如图 6-30 所示，由于 D 关于 x 轴对称，xy 关于 y 是奇函数，故 $I=\iint\limits_D xy\,\mathrm{d}x\,\mathrm{d}y=0$.

（Ⅱ）双纽线$(x^2+y^2)^2=2xy$ 如图 6-31 所示，由于 D 关于原点对称，而$xy=(-x)(-y)$，故

$$I=\iint_D xy\,\mathrm{d}x\,\mathrm{d}y=2\iint_{D_1}xy\,\mathrm{d}x\,\mathrm{d}y\ (D_1\text{ 是 }D\text{ 在第一象限的部分})$$

$$=2\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{\sqrt{\sin 2\theta}}r^3\cos\theta\sin\theta\,\mathrm{d}r=\frac{1}{6}.$$

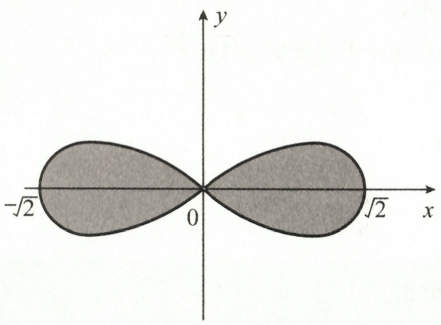

图 6-30

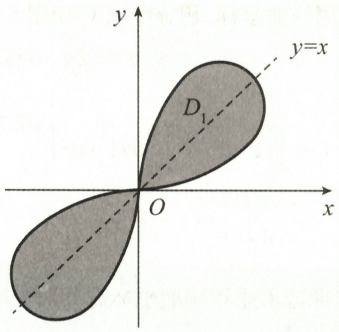

图 6-31

【注】 设 D 关于原点对称，则

$$\iint_D f(x,y)\,\mathrm{d}x\,\mathrm{d}y=\begin{cases}2\iint\limits_{D_1}f(x,y)\,\mathrm{d}x\,\mathrm{d}y, & f(x,y)\text{ 关于 }x,y\text{ 是偶函数},\\ 0, & f(x,y)\text{ 关于 }x,y\text{ 是奇函数},\end{cases}$$

其中 $D_1=D\cap\{(x,y)\mid x\geqslant 0\}$.

(13) 解 将极坐标转化为直角坐标.

D 如图 6-32 阴影部分所示，先对 x 积分后对 y 积分.

$$I=\iint_D(\cos\theta+\sin\theta)r^2\,\mathrm{d}\theta\,\mathrm{d}r$$

$$=\iint_D(x+y)\,\mathrm{d}x\,\mathrm{d}y$$

$$=\int_0^1\mathrm{d}y\int_{1-\sqrt{1-y^2}}^{2-y}(x+y)\,\mathrm{d}x$$

$$=\int_0^1\left(\frac{1}{2}x^2+yx\right)\Big|_{1-\sqrt{1-y^2}}^{2-y}\mathrm{d}y$$

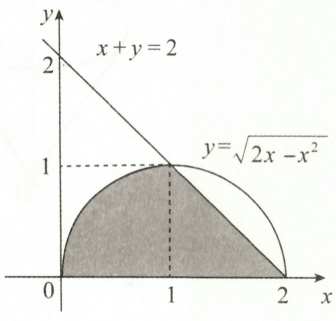

图 6-32

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$$=\int_0^1\left[\frac{(2-y)^2}{2}+y(2-y)-\frac{(1-\sqrt{1-y^2})^2}{2}-y(1-\sqrt{1-y^2})\right]\mathrm{d}y$$

$$=\int_0^1(y\sqrt{1-y^2}+\sqrt{1-y^2}+1-y)\,\mathrm{d}y$$

$$=-\frac{1}{2}\cdot\frac{2}{3}(1-y^2)^{\frac{3}{2}}\Big|_0^1+\frac{1}{4}\pi\cdot1^2+1-\frac{1}{2}y^2\Big|_0^1$$

$$=\frac{1}{3}+\frac{1}{4}\pi+\frac{1}{2}=\frac{5}{6}+\frac{\pi}{4}.$$

【注】 此题选择先对 y 积分后对 x 积分,或采用极坐标计算量都较大.

(14) **解** V 如图 6-33 所示,用球坐标

$$x=r\sin\varphi\cos\theta,\ y=r\sin\varphi\sin\theta,\ z=r\cos\varphi,$$

则 V 为 $0\leqslant r\leqslant R,\ 0\leqslant\varphi\leqslant\dfrac{\pi}{4},\ 0\leqslant\theta\leqslant2\pi$,故

$$I=\iiint_V z\,\mathrm{d}V=\int_0^{2\pi}\mathrm{d}\theta\int_0^{\frac{\pi}{4}}\mathrm{d}\varphi\int_0^R r^2\sin\varphi\cdot r\cos\varphi\,\mathrm{d}r=\int_0^{2\pi}\mathrm{d}\theta\int_0^{\frac{\pi}{4}}\sin\varphi\cos\varphi\,\mathrm{d}\varphi\int_0^R r^3\,\mathrm{d}r$$

$$=2\pi\cdot\left(\frac{1}{2}\sin^2\varphi\right)\Big|_0^{\frac{\pi}{4}}\cdot\left(\frac{1}{4}r^4\right)\Big|_0^R=\frac{\pi R^4}{8}.$$

(15) **解** 用柱面坐标. 积分区域 V 如图 6-34 所示,

$$0\leqslant\theta\leqslant2\pi,\ 0\leqslant r\leqslant1,\ r-1\leqslant z\leqslant\sqrt{1-r^2},$$

故

$$I=\iiint_V z^2\,\mathrm{d}V=\int_0^{2\pi}\mathrm{d}\theta\int_0^1 r\,\mathrm{d}r\int_{r-1}^{\sqrt{1-r^2}}z^2\,\mathrm{d}z$$

$$=\frac{2\pi}{3}\int_0^1\left[r(1-r^2)^{\frac{3}{2}}-r(r-1)^3\right]\mathrm{d}r=\frac{\pi}{6}.$$

【注】 此题若采用球面坐标需分两个区域积分:

$$I=\int_0^{2\pi}\mathrm{d}\theta\int_0^{\frac{\pi}{2}}\mathrm{d}\varphi\int_0^1 r^2\cos^2\varphi\cdot r^2\sin\varphi\,\mathrm{d}r+$$

$$\int_0^{2\pi}\mathrm{d}\theta\int_{\frac{\pi}{2}}^{\pi}\mathrm{d}\varphi\int_0^{\frac{1}{\sin\varphi-\cos\varphi}}r^2\cos^2\varphi\cdot r^2\sin\varphi\,\mathrm{d}r,$$

显然计算量较大.

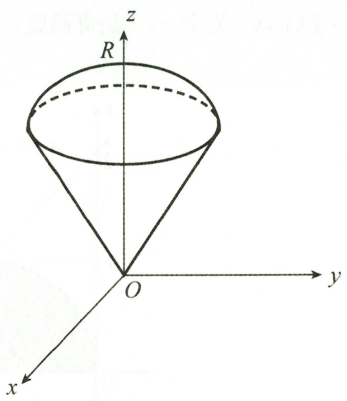

图 6-33

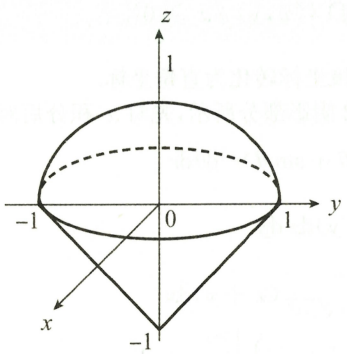

图 6-34

(16) **解** 依题意，立体体积V可视为以$z=\sqrt{5-x^2-y^2}$为曲顶的柱体体积与以$x^2+y^2=4z$为曲顶的柱体体积之差，利用重积分计算. 如图 6-35 所示，两曲面的交线为$\begin{cases}z=\sqrt{5-x^2-y^2},\\ x^2+y^2=4z,\end{cases}$消去$z$，得$D:x^2+y^2=4$，故

$$\begin{aligned}V&=\iint_D\left[\sqrt{5-x^2-y^2}-\frac{1}{4}(x^2+y^2)\right]\mathrm{d}x\,\mathrm{d}y\\&=\int_0^{2\pi}\mathrm{d}\theta\int_0^2\left(\sqrt{5-r^2}-\frac{1}{4}r^2\right)r\,\mathrm{d}r\\&=2\pi\int_0^2\left(\sqrt{5-r^2}-\frac{1}{4}r^2\right)r\,\mathrm{d}r\\&=2\pi\int_0^2 r\sqrt{5-r^2}\,\mathrm{d}r-\frac{\pi}{2}\int_0^2 r^3\mathrm{d}r\\&=\frac{2\pi}{3}(5\sqrt{5}-4).\end{aligned}$$

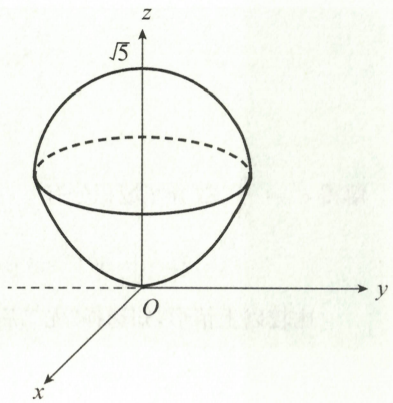

图 6-35

【注】　此题也可采用三重积分计算：$V=\iiint_V\mathrm{d}V$，用柱面坐标，有

$$V=\iiint_V\mathrm{d}V=\int_0^{2\pi}\mathrm{d}\theta\int_0^2 r\,\mathrm{d}r\int_{\frac{1}{4}r^2}^{\sqrt{5-r^2}}\mathrm{d}z=2\pi\int_0^2\left(\sqrt{5-r^2}-\frac{1}{4}r^2\right)r\,\mathrm{d}r=\frac{2\pi}{3}(5\sqrt{5}-4).$$

(17) **解** 如图 6-36 所示，V由两个球面所围.

由$\begin{cases}x^2+y^2+z^2=R^2,\\ x^2+y^2+(z-R)^2=R^2,\end{cases}$解得$z=\frac{R}{2}$，于是积分区域$V$在$xOy$面上的投影曲线为

$$\begin{cases}x^2+y^2=\left(\frac{\sqrt{3}}{2}R\right)^2,\\ z=0.\end{cases}$$

考虑到被积函数仅含z，采用直角坐标的“先二后一”，有

$$\begin{aligned}I&=\iiint_V z^2\mathrm{d}V=\int_0^{\frac{R}{2}}z^2\mathrm{d}z\iint_{x^2+y^2\leqslant 2Rz-z^2}\mathrm{d}x\,\mathrm{d}y+\int_{\frac{R}{2}}^R z^2\mathrm{d}z\iint_{x^2+y^2\leqslant R^2-z^2}\mathrm{d}x\,\mathrm{d}y\\&=\pi\int_0^{\frac{R}{2}}z^2(2Rz-z^2)\mathrm{d}z+\pi\int_{\frac{R}{2}}^R z^2(R^2-z^2)\mathrm{d}z\\&=\pi\left(\frac{R}{2}z^4-\frac{1}{5}z^5\right)\Big|_0^{\frac{R}{2}}+\pi\left(\frac{R^2}{3}z^3-\frac{1}{5}z^5\right)\Big|_{\frac{R}{2}}^R=\frac{59}{480}\pi R^5.\end{aligned}$$

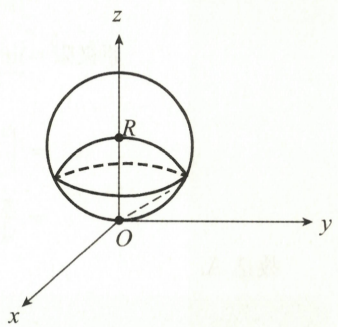

图 6-36

【注】　此题也可采用下列方法：

①“先一后二”，$I=\iint\limits_{x^2+y^2\leqslant\frac{3}{4}R^2}\mathrm{d}x\,\mathrm{d}y\int_{R-\sqrt{R^2-x^2-y^2}}^{\sqrt{R^2-x^2-y^2}}z^2\mathrm{d}z$，显然积分较烦琐.

② 柱面坐标，$I=\int_0^{2\pi}\mathrm{d}\theta\int_0^{\frac{\sqrt{3}}{2}R}r\,\mathrm{d}r\int_{R-\sqrt{R^2-r^2}}^{\sqrt{R^2-r^2}}z^2\mathrm{d}z$.

③ 球面坐标. 由$\begin{cases}z=\frac{R}{2},\\ x^2+y^2=\left(\frac{\sqrt{3}}{2}R\right)^2\end{cases}$化为球坐标，得

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$$\begin{cases} r\cos\varphi=\dfrac{R}{2}, \\ r^2\sin^2\varphi=\left(\dfrac{\sqrt{3}}{2}R\right)^2, \end{cases} \text{即} \begin{cases} r\cos\varphi=\dfrac{R}{2}, \\ r\sin\varphi=\dfrac{\sqrt{3}}{2}R, \end{cases}$$

解得 $\varphi=\dfrac{\pi}{3}$,故分区域积分有

$$I=\int_0^{2\pi}\mathrm{d}\theta\int_0^{\frac{\pi}{3}}\sin\varphi\cos^2\varphi\,\mathrm{d}\varphi\int_0^R r^4\mathrm{d}r+\int_0^{2\pi}\mathrm{d}\theta\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\sin\varphi\cos^2\varphi\,\mathrm{d}\varphi\int_0^{2R\cos\varphi}r^4\mathrm{d}r.$$

比较以上情形,知选择“先二后一”较为方便.

综合题

一、选择题

(1)B.

解 在 $D:0\leqslant x^2+y^2\leqslant 1$ 上有 $\dfrac{\pi}{2}>1\geqslant\sqrt{x^2+y^2}\geqslant x^2+y^2\geqslant(x^2+y^2)^2\geqslant 0$,且 $\cos x$ 在 $\left[0,\dfrac{\pi}{2}\right)$ 上为单调减少函数,故 $0\leqslant\cos\sqrt{x^2+y^2}\leqslant\cos(x^2+y^2)\leqslant\cos(x^2+y^2)^2$,所以 $I_1<I_2<I_3$,选项 B 正确.

(2)A.

解 利用二重积分的定义,有

$$\lim_{n\to\infty}\frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}f\left(\frac{i}{n},\frac{j}{n}\right)=\int_0^1\mathrm{d}x\int_0^1 f(x,y)\mathrm{d}y,$$

$$\text{原极限}=\lim_{n\to\infty}\frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\frac{1}{\left(1+\dfrac{i}{n}\right)\left[1+\left(\dfrac{j}{n}\right)^2\right]}$$

$$=\int_0^1\frac{\mathrm{d}x}{1+x}\int_0^1\frac{\mathrm{d}y}{1+y^2}=\int_0^1\frac{1}{1+x}\cdot\arctan y\Big|_0^1\mathrm{d}x$$

$$=\frac{\pi}{4}\int_0^1\frac{\mathrm{d}x}{1+x}=\frac{\pi}{4}\ln(1+x)\Big|_0^1=\frac{\pi}{4}\ln 2.$$

故选 A.

【注】 一般二次积分的极坐标与直角坐标的相互转化,利用 $\begin{cases}x=r\cos\theta,\\ y=r\sin\theta\end{cases}$ 或 $x^2+y^2=r^2$,$\tan\theta=\dfrac{y}{x}$.

(3)C.

解 由极坐标,可知

$$\int_0^2\mathrm{d}x\int_{\sqrt{3}x}^{x}f(\sqrt{x^2+y^2})\,\mathrm{d}y=-\int_{\frac{\pi}{4}}^{\frac{\pi}{3}}\mathrm{d}\theta\int_0^{2\sec\theta}f(r)r\mathrm{d}r.$$

可排除选项 A,B. 对极坐标交换顺序,有

$$-\int_{\frac{\pi}{4}}^{\frac{\pi}{3}}\mathrm{d}\theta\int_0^{2\sec\theta}f(r)r\mathrm{d}r=-\int_0^{2\sqrt{2}}\mathrm{d}r\int_{\frac{\pi}{4}}^{\frac{\pi}{3}}f(r)r\mathrm{d}\theta-\int_{2\sqrt{2}}^{4}\mathrm{d}r\int_{\arccos\frac{2}{r}}^{\frac{\pi}{3}}f(r)r\mathrm{d}\theta.$$

选项 C 正确.

(4)C.

解 由已知,三角形区域 D 如图 6-37 所示.

记 $A=\iint\limits_D f(x,y)\mathrm{d}x\,\mathrm{d}y$,则 $f(x,y)=xy+A$.

等式两边同时在 D 上积分,得

$$\iint_D f(x,y)\mathrm{d}x\mathrm{d}y=\iint_D xy\mathrm{d}x\mathrm{d}y+\iint_D A\mathrm{d}x\mathrm{d}y.$$

故 $A=\iint_D xy\mathrm{d}x\mathrm{d}y+2A$.

作辅助线 $y=-x$，将 D 分成 D_1 与 D_2，D_3 与 D_4，则

$$\iint_D xy\mathrm{d}x\mathrm{d}y=\iint_{D_1\cup D_2} xy\mathrm{d}x\mathrm{d}y+\iint_{D_3\cup D_4} xy\mathrm{d}x\mathrm{d}y.$$

$D_1\cup D_2$ 关于 y 轴对称，xy 关于 x 是奇函数，$D_3\cup D_4$ 关于 x 轴对称，xy 关于 y 是奇函数，故 $\iint_D xy\mathrm{d}x\mathrm{d}y=0$，从而 $A=0+2A$，即 $A=0$. 故 $f(x,y)=xy=yx$. 选项 C 正确.

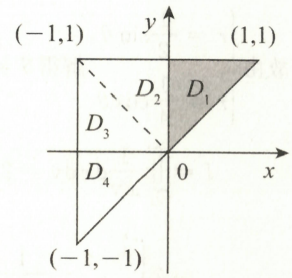

图 6-37

选项 A,B,D 中等号左端均为零，但右端不为零，故排除选项 A,B,D.

(5)C.

解 由二重积分中值定理，有

$$\iint_{D_t}(\sin x+\cos y)\mathrm{d}x\mathrm{d}y=(\sin\xi+\cos\eta)\iint_{D_t}\mathrm{d}x\mathrm{d}y,$$

其中，$(\xi,\eta)\in D_t$. 当 $t\to0^+$ 时，$\sin\xi+\cos\eta\to1$.

故

$$\begin{aligned}
&\lim_{t\to0^+}\frac{1}{t^2}\iint_{D_t}(\sin x+\cos y)\mathrm{d}x\mathrm{d}y\\
&=\lim_{t\to0^+}\frac{1}{t^2}\iint_{D_t}\mathrm{d}x\mathrm{d}y\\
&=\lim_{t\to0^+}\frac{1}{t^2}\cdot4\int_0^t y(x)\mathrm{d}x\\
&=4\lim_{t\to0^+}\frac{1}{t^2}\int_{\frac{\pi}{2}}^{0}t\sin^3\theta\cdot3t\cos^2\theta(-\sin\theta)\mathrm{d}\theta\\
&=12\lim_{t\to0^+}\frac{t^2}{t^2}\int_0^{\frac{\pi}{2}}(\sin^4\theta-\sin^6\theta)\,\mathrm{d}\theta\\
&=12\cdot\left[\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}\cdot\left(1-\frac{5}{6}\right)\right]=\frac{3}{8}\pi.
\end{aligned}$$

选项 C 正确.

二、填空题

(1)4π.

解 积分区域 D 如图 6-38 所示，为去掉绝对值符号，用 $y=x+\pi$ 将 D 划分为 D_1 与 D_2，

$$D_1:-\pi\leqslant x-y\leqslant0,D_2:-2\pi\leqslant x-y\leqslant-\pi,$$

故

$$\begin{aligned}
I&=\iint_D|\sin(x-y)|\mathrm{d}x\mathrm{d}y\\
&=\iint_{D_1}[-\sin(x-y)]\mathrm{d}x\mathrm{d}y+\iint_{D_2}\sin(x-y)\mathrm{d}x\mathrm{d}y\\
&=\int_0^{\pi}\mathrm{d}y\int_0^{y}[-\sin(x-y)]\mathrm{d}x+\int_{\pi}^{2\pi}\mathrm{d}y\int_{y-\pi}^{y}[-\sin(x-y)]\mathrm{d}x+\\
&\quad\int_{\pi}^{2\pi}\mathrm{d}y\int_0^{y-\pi}\sin(x-y)\mathrm{d}x\\
&=4\pi.
\end{aligned}$$

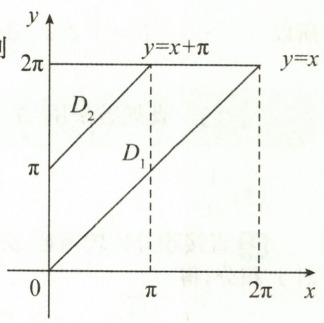

图 6-38

(2)$(\ln2)^2$.

解 利用极坐标求解. 如图 6-39 所示，D 关于 $y=x$ 对称，且

$$D:\frac{1}{4}\cos\theta\leqslant r\leqslant\frac{1}{2}\cos\theta,\ \frac{1}{4}\sin\theta\leqslant r\leqslant\frac{1}{2}\sin\theta,$$

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故由$\begin{cases} r=\dfrac{1}{2}\sin\theta, \\ r=\dfrac{1}{4}\cos\theta \end{cases}$解得$\theta=\arctan\dfrac{1}{2}$,故

$$I=\iint_D\frac{1}{xy}\mathrm{d}x\,\mathrm{d}y=2\int_{\arctan\frac{1}{2}}^{\frac{\pi}{4}}\mathrm{d}\theta\int_{\frac{1}{4}\cos\theta}^{\frac{1}{2}\sin\theta}\frac{\mathrm{d}r}{r\sin\theta\cos\theta}$$

$$=2\int_{\arctan\frac{1}{2}}^{\frac{\pi}{4}}\frac{1}{\sin\theta\cos\theta}\ln\frac{\frac{1}{2}\sin\theta}{\frac{1}{4}\cos\theta}\mathrm{d}\theta$$

$$=2\int_{\arctan\frac{1}{2}}^{\frac{\pi}{4}}\frac{1}{\tan\theta}\ln(2\tan\theta)\mathrm{d}(\tan\theta)$$

$$=(\ln 2)^2.$$

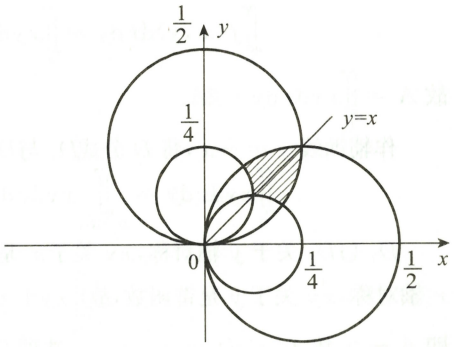

图 6-39

(3) $\dfrac{1}{4}(1-\cos 1)$.

解 交换积分顺序,得

$$I=\int_0^1\mathrm{d}x\int_{\sqrt{x}}^1 y\sin(1-x)^2\mathrm{d}y=\int_0^1\frac{y^2}{2}\sin(1-x)^2\Big|_{\sqrt{x}}^1\mathrm{d}x$$

$$=\frac{1}{2}\int_0^1(1-x)\sin(1-x)^2\mathrm{d}x=-\frac{1}{4}\int_0^1\sin(1-x)^2\mathrm{d}[(1-x)^2]$$

$$=\frac{1}{4}\cos(1-x)^2\Big|_0^1=\frac{1}{4}(1-\cos 1).$$

(4) $2\pi\mathrm{e}^2$.

解 积分区域V如图 6-40 所示,考虑到被积函数,可采用直角坐标下的"先二后一",则

$$I=\int_1^2\mathrm{e}^z\mathrm{d}z\iint_{D_z}\frac{1}{\sqrt{x^2+y^2}}\mathrm{d}x\,\mathrm{d}y,$$

其中$D_z:x^2+y^2\leqslant z^2$,又因为

$$\iint_{D_z}\frac{1}{\sqrt{x^2+y^2}}\mathrm{d}x\,\mathrm{d}y=\int_0^{2\pi}\mathrm{d}\theta\int_0^z\frac{1}{r}r\,\mathrm{d}r=2\pi z,$$

所以 $$I=\int_1^2 2\pi z\mathrm{e}^z\mathrm{d}z=2\pi\left(z\mathrm{e}^z\Big|_1^2-\int_1^2\mathrm{e}^z\mathrm{d}z\right)=2\pi\mathrm{e}^2.$$

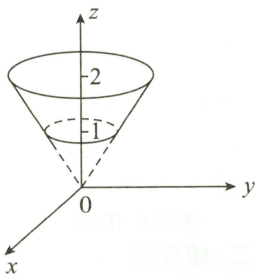

图 6-40

【注】 此题若采用"先一后二"或球面坐标的方法,计算量较大.

(5) $\dfrac{1}{4\mathrm{e}}$.

解 直接积分比较困难,交换积分顺序,积分区域如图 6-41 所示,先对x积分,得

$$I=\iiint_V(1-y)\mathrm{e}^{-(1-y-z)^2}\mathrm{d}V$$

$$=\iint_{D_{yz}}\mathrm{d}y\,\mathrm{d}z\int_0^{1-y-z}(1-y)\mathrm{e}^{-(1-y-z)^2}\mathrm{d}x$$

$$=\iint_{D_{yz}}(1-y)(1-y-z)\mathrm{e}^{-(1-y-z)^2}\mathrm{d}y\mathrm{d}z$$

$$=\int_0^1(1-y)\mathrm{d}y\int_0^{1-y}(1-y-z)\mathrm{e}^{-(1-y-z)^2}\mathrm{d}z$$

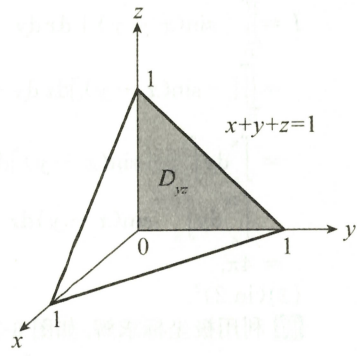

图 6-41

$$=\frac{1}{2}\int_0^1(1-y)\left[\mathrm{e}^{-(1-y-z)^2}\right]\Big|_0^{1-y}\mathrm{d}y$$
$$=\frac{1}{2}\int_0^1(1-y)\left[1-\mathrm{e}^{-(1-y)^2}\right]\mathrm{d}y=\frac{1}{4\mathrm{e}}.$$

(6)9π.

解 积分区域V如图 6-42 所示，由于被积函数仅与z有关，所以采用“先二后一”. D_z为$x^2+y^2\leqslant 1+2z^2(1\leqslant z\leqslant 2)$，$D_z$的面积为$\pi(1+2z^2)$，故

$$I=\iiint_V z\,\mathrm{d}V=\int_1^2 z\,\mathrm{d}z\iint_{D_z}\mathrm{d}x\,\mathrm{d}y=\int_1^2\pi(1+2z^2)z\,\mathrm{d}z=9\pi.$$

(7)$\dfrac{\pi}{8}$.

解 V由下半圆锥与上半球面所围，如图 6-43 所示，V关于yOz面对称，且被积函数x是奇函数，故$\iiint_V x\,\mathrm{d}V=0$，则

$$I=\iiint_V(x+z)\mathrm{d}V=0+\iiint_V z\,\mathrm{d}V=\int_0^{2\pi}\mathrm{d}\theta\int_0^{\frac{\pi}{4}}\mathrm{d}\varphi\int_0^1 r\cos\varphi\cdot r^2\sin\varphi\,\mathrm{d}r=\frac{\pi}{8}.$$

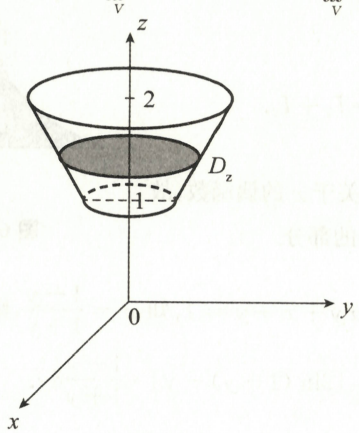

图 6-42

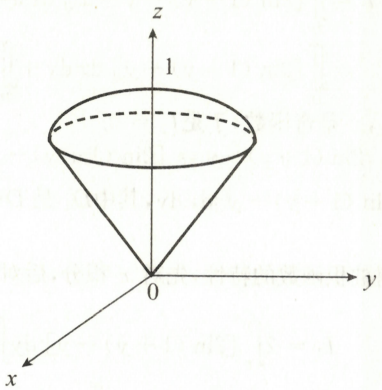

图 6-43

三、解答题

(1) **解** 当$y+1=\dfrac{2}{1+x}$时，双曲线$y+1=\dfrac{2}{1+x}$将D分成D_1与D_2，如图 6-44 所示.

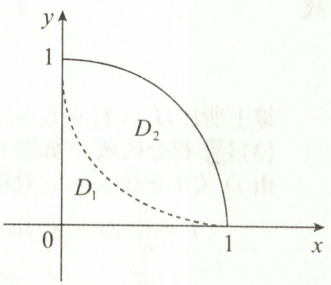

图 6-44

$$\max\left\{y+1,\frac{2}{1+x}\right\}=\begin{cases}\dfrac{2}{1+x}, & (x,y)\in D_1,\\ 1+y, & (x,y)\in D_2,\end{cases}$$

故$I=\iint_{D_1}\dfrac{2}{1+x}\mathrm{d}x\,\mathrm{d}y+\iint_{D_2}(1+y)\mathrm{d}x\,\mathrm{d}y\overset{\text{记}}{=\!=}I_1+I_2$.

$$I_1=\iint_{D_1}\frac{2}{1+x}\mathrm{d}x\,\mathrm{d}y=\int_0^1\mathrm{d}x\int_0^{\frac{1-x}{1+x}}\frac{2}{1+x}\mathrm{d}y$$
$$=\int_0^1\left(\frac{2}{1+x}\cdot\frac{1-x}{1+x}\right)\mathrm{d}x=2\int_0^1\frac{1-x}{(1+x)^2}\mathrm{d}x$$
$$=2\int_0^1\left[\frac{-1}{1+x}+\frac{2}{(1+x)^2}\right]\mathrm{d}x=-2\ln(1+x)\Big|_0^1+4\int_0^1\frac{\mathrm{d}(1+x)}{(1+x)^2}$$
$$=-2(\ln 2-0)-4\frac{1}{1+x}\Big|_0^1=-2\ln 2-4\left(\frac{1}{2}-1\right)$$
$$=2-2\ln 2,$$

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$$I_2=\iint_{D_2}(1+y)\mathrm{d}x\,\mathrm{d}y=\int_0^1\mathrm{d}y\int_{\frac{1-y}{1+y}}^{\sqrt{1-y^2}}(1+y)\mathrm{d}x$$
$$=\int_0^1(1+y)\left(\sqrt{1-y^2}-\frac{1-y}{1+y}\right)\mathrm{d}y$$
$$=\int_0^1\sqrt{1-y^2}\,\mathrm{d}y+\int_0^1 y\sqrt{1-y^2}\,\mathrm{d}y-\int_0^1(1-y)\mathrm{d}y$$
$$=\frac{\pi}{4}-\frac{1}{2}\int_0^1\sqrt{1-y^2}\,\mathrm{d}(1-y^2)-\left(y-\frac{1}{2}y^2\right)\Big|_0^1$$
$$=\frac{\pi}{4}-\frac{1}{2}\cdot\frac{2}{3}(1-y^2)^{\frac{3}{2}}\Big|_0^1-\frac{1}{2}=\frac{\pi}{4}-\frac{1}{6},$$

故 $I=I_1+I_2=2-2\ln 2+\dfrac{\pi}{4}-\dfrac{1}{6}=\dfrac{\pi}{4}-2\ln 2+\dfrac{11}{6}$.

(2)解 以 $-x$ 代替 x,方程 $|x|y+|x|+y=1$ 不变,可知曲线关于 y 轴对称.

当 $x\geqslant 0$ 时,由 $xy+x+y=1$,得 $y=\dfrac{1-x}{1+x}$,即 $y+1=\dfrac{2}{1+x}$,它是以 $x=-1$,$y=-1$ 为渐近线的双曲线. D 如图 6-45 阴影部分所示.

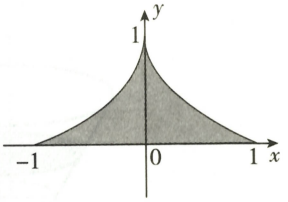

图 6-45

$$I=\iint_D[2\ln(1+y)-y+x]\,\mathrm{d}x\,\mathrm{d}y$$
$$=\iint_D[2\ln(1+y)-y]\,\mathrm{d}x\,\mathrm{d}y+\iint_D x\,\mathrm{d}x\,\mathrm{d}y\xlongequal{\text{记}}I_1+I_2.$$

对于 I_2,x 是奇函数,于是 $I_2=0$.

对于 I_1,$2\ln(1+y)-y=[2\ln(1+y)-y]x^0$,它是关于 x 的偶函数,故 $I_1=2\iint_{D_1}[2\ln(1+y)-y]\mathrm{d}x\,\mathrm{d}y$,其中 D_1 是 D 在第一象限的部分.

考虑到被积函数的特性,先对 x 积分,后对 y 积分,由 $xy+x+y=1$,知 $x=\dfrac{1-y}{1+y}$,则

$$I_1=2\int_0^1[2\ln(1+y)-y]\,\mathrm{d}y\int_0^{\frac{1-y}{1+y}}\mathrm{d}x=2\int_0^1[2\ln(1+y)-y]\cdot\frac{1-y}{1+y}\mathrm{d}y.$$

注意到 $[2\ln(1+y)-y]'=\dfrac{2}{1+y}-1=\dfrac{1-y}{1+y}$,

故
$$I_1=2\int_0^1[2\ln(1+y)-y]\,\mathrm{d}[2\ln(1+y)-y]$$
$$=2\times\frac{1}{2}[2\ln(1+y)-y]^2\Big|_0^1=(2\ln 2-1)^2.$$

综上所述,$I=I_1+I_2=(2\ln 2-1)^2$.

(3)解 积分区域 D 如图 6-46 所示.

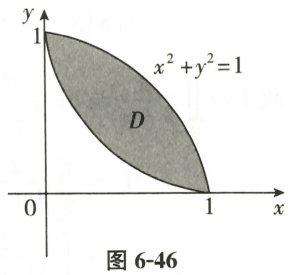

图 6-46

由 D 关于直线 $y=x$ 对称,知

$$I=\iint_D(2x-y^2)\mathrm{d}x\,\mathrm{d}y=\iint_D(2y-x^2)\mathrm{d}x\,\mathrm{d}y$$
$$=\iint_D 2y\,\mathrm{d}x\,\mathrm{d}y-\iint_D x^2\mathrm{d}x\,\mathrm{d}y\xlongequal{\text{记}}I_1-I_2,$$
$$I_1=\iint_D 2y\,\mathrm{d}x\,\mathrm{d}y=\int_0^1\mathrm{d}x\int_{1-\sqrt{2x-x^2}}^{\sqrt{1-x^2}}2y\,\mathrm{d}y$$
$$=\int_0^1\left[(1-x^2)-\left(1-\sqrt{2x-x^2}\right)^2\right]\mathrm{d}x$$
$$=\int_0^1(2\sqrt{2x-x^2}-2x)\mathrm{d}x=-1+2\int_0^1\sqrt{2x-x^2}\,\mathrm{d}x,$$

而
$$2\int_0^1\sqrt{2x-x^2}\,\mathrm{d}x=2\int_0^1\sqrt{1-(x-1)^2}\,\mathrm{d}(x-1)\xlongequal{x-1=t}2\int_{-1}^0\sqrt{1-t^2}\,\mathrm{d}t$$

$$=2\cdot\frac{\pi}{4}\cdot1^2=\frac{\pi}{2},$$

故$I_1=-1+\frac{\pi}{2}$.

$$I_2=\iint_D x^2\,dx\,dy=\int_0^1 dx\int_{1-\sqrt{2x-x^2}}^{\sqrt{1-x^2}}x^2\,dy$$
$$=\int_0^1 x^2\left[\sqrt{1-x^2}-\left(1-\sqrt{2x-x^2}\right)\right]dx$$
$$=\int_0^1 x^2\sqrt{1-x^2}\,dx-\int_0^1 x^2\,dx+\int_0^1 x^2\sqrt{2x-x^2}\,dx,$$

又
$$\int_0^1 x^2\sqrt{1-x^2}\,dx\xlongequal{x=\sin t}\int_0^{\frac{\pi}{2}}\sin^2t\cos^2t\,dt$$
$$=\int_0^{\frac{\pi}{2}}\sin^2t(1-\sin^2t)\,dt=\int_0^{\frac{\pi}{2}}\sin^2t\,dt-\int_0^{\frac{\pi}{2}}\sin^4t\,dt$$
$$=\frac{1}{2}\cdot\frac{\pi}{2}-\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}=\frac{\pi}{16},$$
$$\int_0^1 x^2\,dx=\frac{1}{3},$$
$$\int_0^1 x^2\sqrt{2x-x^2}\,dx=\int_0^1 x^2\sqrt{1-(x-1)^2}\,dx$$
$$\xlongequal{x-1=\sin t}\int_{-\frac{\pi}{2}}^0(1+\sin t)^2\cos^2t\,dt$$
$$\xlongequal{t=-u}\int_0^{\frac{\pi}{2}}(1-\sin u)^2\cos^2u\,du$$
$$=\int_0^{\frac{\pi}{2}}(1-2\sin u+\sin^2u)\cos^2u\,du$$
$$=\int_0^{\frac{\pi}{2}}\cos^2u\,du-2\int_0^{\frac{\pi}{2}}\sin u\cos^2u\,du+\int_0^{\frac{\pi}{2}}\sin^2u\cos^2u\,du$$
$$=\frac{1}{2}\cdot\frac{\pi}{2}+2\int_0^{\frac{\pi}{2}}\cos^2u\,d(\cos u)+\frac{\pi}{16}$$
$$=\frac{\pi}{4}+2\cdot\frac{1}{3}\cos^3u\Big|_0^{\frac{\pi}{2}}+\frac{\pi}{16}=-\frac{2}{3}+\frac{5\pi}{16},$$

故$I=-1+\frac{\pi}{2}-\left(\frac{\pi}{16}-\frac{1}{3}-\frac{2}{3}+\frac{5\pi}{16}\right)=\frac{\pi}{8}$.

(4) 解 已知曲线$(x+y)^3=xy$过原点$(0,0)$,其在极坐标下化为

$$r(\theta)=\frac{\sin\theta\cos\theta}{(\sin\theta+\cos\theta)^3}=\frac{1}{2}\frac{\sin2\theta}{(\sin\theta+\cos\theta)^3}.$$

当$r(\theta)=0$时,$\sin2\theta=0$,得$\theta=0,\theta=\frac{\pi}{2}$.

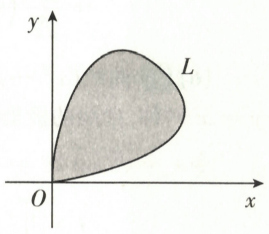

图 6-47

又$(x+y)^3=xy$中x与y轮换后,方程不变,知该曲线关于直线$y=x$对称,如图 6-47 所示,则

$$\iint_D(x-y)^3\,dx\,dy=\frac{1}{2}\iint_D[(x-y)^3+(y-x)^3]\,dx\,dy=0.$$

故
$$I=\iint_D[(x-y)^3+1]\,dx\,dy$$
$$=\iint_D dx\,dy=\int_0^{\frac{\pi}{2}}d\theta\int_0^{r(\theta)}r\,dr$$
$$=\int_0^{\frac{\pi}{2}}\frac{1}{2}r^2\Big|_0^{r(\theta)}d\theta=\frac{1}{2}\int_0^{\frac{\pi}{2}}\frac{\sin^2\theta\cos^2\theta}{(\sin\theta+\cos\theta)^6}d\theta$$

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$$=\frac{1}{2}\int_0^{\frac{\pi}{2}}\frac{\sin^2\theta\cos^2\theta}{\cos^6\theta(1+\tan\theta)^6}\mathrm{d}\theta=\frac{1}{2}\int_0^{\frac{\pi}{2}}\frac{\tan^2\theta}{(1+\tan\theta)^6}\mathrm{d}(\tan\theta)$$

$$\xlongequal{1+\tan\theta=t}\frac{1}{2}\int_1^{+\infty}\frac{(t-1)^2}{t^6}\mathrm{d}t=\frac{1}{2}\left(-\frac{1}{3t^3}+\frac{1}{2t^4}-\frac{1}{5t^5}\right)\Big|_1^{+\infty}=\frac{1}{60}.$$

(5) **解** 在方程 $x^2+(y-|x|)^2=1$ 与 $x^2+(y-|x|)^2=2$ 中,以 $-x$ 代替 x,方程不变,可知两条曲线均关于 y 轴对称.

$$I=\iint_D\frac{x+1}{2x^2+y^2}\mathrm{d}x\,\mathrm{d}y=\iint_D\frac{x}{2x^2+y^2}\mathrm{d}x\,\mathrm{d}y+\iint_D\frac{1}{2x^2+y^2}\mathrm{d}x\,\mathrm{d}y$$

$$\xlongequal{记}I_1+I_2.$$

对于 I_1,由 $\frac{x}{2x^2+y^2}$ 关于 x 是奇函数,D 关于 y 轴对称,知

$$I_1=\iint_D\frac{x}{2x^2+y^2}\mathrm{d}x\,\mathrm{d}y=0.$$

对于 I_2,由 $\frac{1}{2x^2+y^2}$ 关于 x 是偶函数,D 关于 y 轴对称,知

$$I_2=\iint_D\frac{1}{2x^2+y^2}\mathrm{d}x\,\mathrm{d}y=2\iint_{D_1}\frac{1}{2x^2+y^2}\mathrm{d}x\,\mathrm{d}y.$$

其中,D_1 是 D 在第一象限内的阴影部分,如图 6-48 所示.

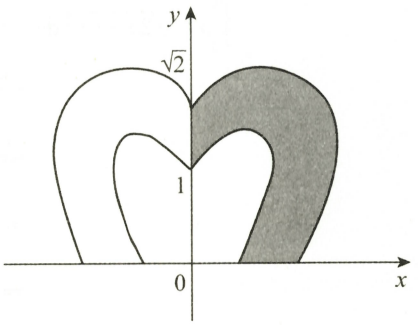

图 6-48

当 $x\geqslant 0$ 时,$x^2+(y-x)^2=1$,即 $2x^2+y^2-2xy=1$,其在极坐标下化为

$$r=\frac{1}{\sqrt{1+\cos^2\theta-\sin 2\theta}}.$$

同样,当 $x\geqslant 0$ 时,$x^2+(y-x)^2=2$ 在极坐标下化为 $r=\frac{\sqrt{2}}{\sqrt{1+\cos^2\theta-\sin 2\theta}}$,则

$$I=2\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_{\frac{1}{\sqrt{1+\cos^2\theta-\sin 2\theta}}}^{\frac{\sqrt{2}}{\sqrt{1+\cos^2\theta-\sin 2\theta}}}\frac{r\mathrm{d}r}{2r^2\cos^2\theta+r^2\sin^2\theta}$$

$$=2\cdot\frac{1}{2}\int_0^{\frac{\pi}{2}}\frac{1}{2\cos^2\theta+\sin^2\theta}\cdot\ln r^2\Big|_{\frac{1}{\sqrt{1+\cos^2\theta-\sin 2\theta}}}^{\frac{\sqrt{2}}{\sqrt{1+\cos^2\theta-\sin 2\theta}}}\mathrm{d}\theta$$

$$=\ln 2\int_0^{\frac{\pi}{2}}\frac{\mathrm{d}\theta}{2\cos^2\theta+\sin^2\theta}=\ln 2\int_0^{\frac{\pi}{2}}\frac{\mathrm{d}\theta}{\cos^2\theta(2+\tan^2\theta)}=\ln 2\int_0^{\frac{\pi}{2}}\frac{\mathrm{d}(\tan\theta)}{2+\tan^2\theta}$$

$$\xlongequal{\tan\theta=t}\ln 2\int_0^{+\infty}\frac{\mathrm{d}t}{2+t^2}=\frac{1}{\sqrt{2}}\ln 2\arctan\frac{t}{\sqrt{2}}\Big|_0^{+\infty}$$

$$=\frac{\ln 2}{\sqrt{2}}\left(\frac{\pi}{2}-0\right)=\frac{\sqrt{2}\pi\ln 2}{4}.$$

(6) **解** 曲线 $(x^2+y^2)^{\frac{3}{2}}=2\sqrt{2}xy$ 过点 $(0,0)$ 与点 $(1,1)$. 由 x 与 y 互换后方程不变,知 D 关于直线 $y=x$ 对称. 曲线的极坐标方程为 $r=\sqrt{2}\sin 2\theta$.

令 $r=\sqrt{2}\sin 2\theta=0$,得 $\theta=0,\theta=\frac{\pi}{2}$.

$$I=\iint_D\left[(x-y)+2\sqrt{xy}\right]^2\mathrm{d}x\,\mathrm{d}y$$

$$=\iint_D\left[(x-y)^2+4xy\right]\mathrm{d}x\,\mathrm{d}y+\iint_D 4\sqrt{xy}(x-y)\mathrm{d}x\,\mathrm{d}y$$

$$=\iint_D(x+y)^2\mathrm{d}x\,\mathrm{d}y+4\iint_D\sqrt{xy}(x-y)\mathrm{d}x\,\mathrm{d}y\xlongequal{记}I_1+I_2.$$

由轮换对称性,知

$$I_2=\frac{1}{2}\iint_D\left[4\sqrt{xy}(x-y)+4\sqrt{yx}(y-x)\right]\mathrm{d}x\,\mathrm{d}y=0.$$

$$\begin{aligned}I_1&=\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{\sqrt{2}\sin 2\theta}(r\cos\theta+r\sin\theta)^2\cdot r\mathrm{d}r\\&=\int_0^{\frac{\pi}{2}}(\cos\theta+\sin\theta)^2\cdot\frac{1}{4}r^4\Big|_0^{\sqrt{2}\sin 2\theta}\mathrm{d}\theta\\&=\frac{1}{4}\int_0^{\frac{\pi}{2}}(1+\sin 2\theta)(\sqrt{2})^4\sin^4 2\theta\,\mathrm{d}\theta\\&=\int_0^{\frac{\pi}{2}}\sin^4 2\theta\,\mathrm{d}\theta+\int_0^{\frac{\pi}{2}}\sin^5 2\theta\,\mathrm{d}\theta\\&\xlongequal{2\theta=t}\int_0^{\pi}\sin^4 t\cdot\frac{1}{2}\mathrm{d}t+\int_0^{\pi}\sin^5 t\cdot\frac{1}{2}\mathrm{d}t\\&=\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}+\frac{4}{5}\cdot\frac{2}{3}\\&=\frac{3\pi}{16}+\frac{8}{15},\end{aligned}$$

故 $I=\dfrac{3\pi}{16}+\dfrac{8}{15}$.

(7) 解　$I=\iint_D(x^2+xy-y^2)\mathrm{d}x\,\mathrm{d}y=\iint_D xy\,\mathrm{d}x\,\mathrm{d}y+\iint_D(x^2-y^2)\mathrm{d}x\,\mathrm{d}y.$

依题设，D 关于直线 $y=x$ 对称，由轮换对称性，有

$$\iint_D(x^2-y^2)\mathrm{d}x\,\mathrm{d}y=\frac{1}{2}\iint_D(x^2-y^2+y^2-x^2)\,\mathrm{d}x\,\mathrm{d}y=0.$$

故

$$\begin{aligned}I&=\iint_D xy\,\mathrm{d}x\,\mathrm{d}y=\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{\frac{1}{\sqrt[4]{\sin^4\theta+\cos^4\theta}}}r\cos\theta r\sin\theta r\mathrm{d}r\\&=\int_0^{\frac{\pi}{2}}\cos\theta\sin\theta\cdot\frac{1}{4}r^4\Big|_0^{\frac{1}{\sqrt[4]{\sin^4\theta+\cos^4\theta}}}\mathrm{d}\theta\\&=\frac{1}{4}\int_0^{\frac{\pi}{2}}\frac{\cos\theta\sin\theta}{\sin^4\theta+\cos^4\theta}\mathrm{d}\theta\\&\xlongequal{\sin\theta=t}\frac{1}{4}\int_0^1\frac{t\,\mathrm{d}t}{t^4+(1-t^2)^2}=\frac{1}{4}\cdot\frac{1}{2}\int_0^1\frac{\mathrm{d}(t^2)}{2t^4-2t^2+1}\\&\xlongequal{t^2=u}\frac{1}{8}\int_0^1\frac{\mathrm{d}u}{2u^2-2u+1}=\frac{1}{16}\int_0^1\frac{\mathrm{d}u}{u^2-u+\frac{1}{2}}\\&=\frac{1}{16}\int_0^1\frac{\mathrm{d}\left(u-\frac{1}{2}\right)}{\left(u-\frac{1}{2}\right)^2+\left(\frac{1}{2}\right)^2}=\frac{1}{16}\times 2\arctan\frac{u-\frac{1}{2}}{\frac{1}{2}}\Big|_0^1\\&=\frac{1}{8}\cdot[\arctan 1-\arctan(-1)]=\frac{\pi}{16}.\end{aligned}$$

(8) 解　积分区域 D 如图 6-49 所示，显然 D 关于直线 $x=y$ 对称，则

$$\begin{aligned}I&=\iint_D\mathrm{e}^{(x+y)^2}(\sin^2 x+\cos^2 y)\mathrm{d}x\,\mathrm{d}y\\&=\frac{1}{2}\iint_D[\mathrm{e}^{(x+y)^2}(\sin^2 x+\cos^2 y)+\mathrm{e}^{(y+x)^2}(\sin^2 y+\cos^2 x)]\mathrm{d}x\,\mathrm{d}y\\&=\iint_D\mathrm{e}^{(x+y)^2}\mathrm{d}x\,\mathrm{d}y=\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_{\frac{1}{\cos\theta+\sin\theta}}^{\frac{2}{\cos\theta+\sin\theta}}\mathrm{e}^{r^2(\cos\theta+\sin\theta)^2}\cdot r\mathrm{d}r\\&=\frac{1}{2}\int_0^{\frac{\pi}{2}}\frac{1}{(\cos\theta+\sin\theta)^2}\cdot\mathrm{e}^{r^2(\cos\theta+\sin\theta)^2}\Big|_{\frac{1}{\cos\theta+\sin\theta}}^{\frac{2}{\cos\theta+\sin\theta}}\mathrm{d}\theta\end{aligned}$$

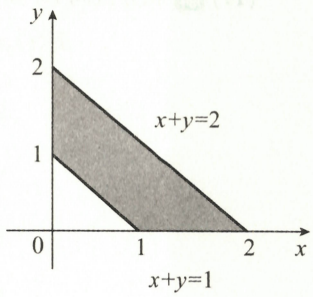

图 6-49

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$$=\frac{1}{2}(e^4-e)\int_0^{\frac{\pi}{2}}\frac{d\theta}{(\cos\theta+\sin\theta)^2}=\frac{1}{2}(e^4-e)\int_0^{\frac{\pi}{2}}\frac{d(\tan\theta)}{(1+\tan\theta)^2}$$

$$=\frac{1}{2}(e^4-e)\left(-\frac{1}{1+\tan\theta}\right)\Big|_0^{\frac{\pi}{2}}=\frac{1}{2}e(e^3-1)$$

(9) **解** 积分区域 D 如图 6-50 阴影部分所示,D 关于 y 轴对称,被积函数关于 x 是偶函数,D_1 为 D 位于 y 轴右侧的部分,下面采用极坐标:

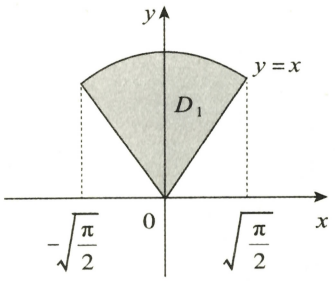

图 6-50

$$I=2\iint_{D_1}e^{-(x^2+y^2)}\sin(x^2+y^2)\,dx\,dy$$

$$=2\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}d\theta\int_0^{\sqrt{\pi}}re^{-r^2}\sin r^2\,dr$$

$$=2\left(\frac{\pi}{2}-\frac{\pi}{4}\right)\cdot\frac{1}{2}\int_0^{\sqrt{\pi}}e^{-r^2}\sin r^2\,d(r^2)$$

$$\xlongequal{r^2=t}\frac{\pi}{4}\int_0^{\pi}e^{-t}\sin t\,dt.$$

记 $A=\int_0^{\pi}e^{-t}\sin t\,dt$,则

$$A=-\int_0^{\pi}\sin t\,d(e^{-t})=-\left(e^{-t}\sin t\Big|_0^{\pi}-\int_0^{\pi}e^{-t}\cos t\,dt\right)$$

$$=-\int_0^{\pi}\cos t\,d(e^{-t})=-\left(e^{-t}\cos t\Big|_0^{\pi}+\int_0^{\pi}e^{-t}\sin t\,dt\right)$$

$$=e^{-\pi}+1-A,$$

故 $A=\frac{1}{2}(1+e^{-\pi})$,所以 $I=\frac{\pi}{4}\cdot\frac{1}{2}(1+e^{-\pi})=\frac{\pi}{8}(1+e^{-\pi})$.

(10) **解** $\frac{0}{0}$ 型,利用洛必达法则,需交换积分顺序,二次积分表示的积分区域如图 6-51 所示.

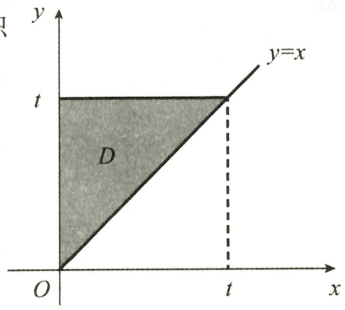

图 6-51

$$\int_0^t dx\int_x^t\sin(xy)^2dy=\int_0^t dy\int_0^y\sin(xy)^2dx,$$

故

$$\text{原式}=\lim_{t\to0^+}\frac{\int_0^t dy\int_0^y\sin(xy)^2dx}{t^6}$$

$$=\lim_{t\to0^+}\frac{\int_0^t\sin(tx)^2dx}{6t^5}\xlongequal[x=\frac{1}{t}u]{tx=u}\lim_{t\to0^+}\frac{\int_0^{t^2}\sin u^2\cdot\frac{1}{t}du}{6t^5}$$

$$=\lim_{t\to0^+}\frac{\int_0^{t^2}\sin u^2du}{6t^6}=\lim_{t\to0^+}\frac{2t\sin t^4}{36t^5}=\frac{1}{18}.$$

(11) **解** 积分区间 D_t 如图 6-52 所示,故

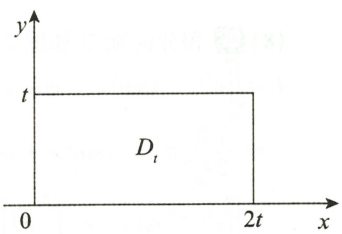

图 6-52

$$F(t)=\iint_{D_t}f''_{yx}(x,y)\,dx\,dy$$

$$=\int_0^{2t}dx\int_0^t f''_{yx}(x,y)\,dy$$

$$=\int_0^{2t}dx\int_0^t d[f'_x(x,y)]$$

$$=\int_0^{2t}f'_x(x,y)\Big|_0^t\,dx$$

$$=\int_0^{2t}[f'_x(x,t)-f'_x(x,0)]\,dx$$

$$=\int_0^{2t}\mathrm{d}[f(x,t)]-\int_0^{2t}\mathrm{d}[f(x,0)]$$
$$=f(x,t)\Big|_0^{2t}-f(x,0)\Big|_0^{2t}$$
$$=f(2t,t)-f(0,t)-[f(2t,0)-f(0,0)],$$
$$\lim_{t\to0^+}\frac{F(t)}{t}=\lim_{t\to0^+}\frac{f(2t,t)-f(2t,0)-[f(0,t)-f(0,0)]}{t}$$
$$\overset{\text{洛必达}}{\underset{\text{法则}}{=}}\lim_{t\to0^+}\frac{f_1'(2t,t)\cdot2+f_2'(2t,t)\cdot1-f_1'(2t,0)\cdot2}{1}-\lim_{t\to0^+}\frac{f(0,t)-f(0,0)}{t}$$
$$=2f_1'(0,0)+f_2'(0,0)-2f_1'(0,0)-f_2'(0,0)=0.$$

(12) 解法1 转化为直角坐标，积分区域 D 如图 6-53 所示.

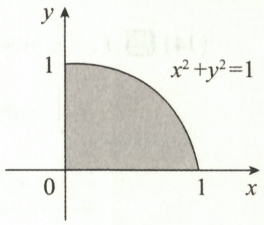

图 6-53

$$I=\iint_D 2r\cos\theta r\sin\theta\,\mathrm{e}^{r^2(\cos^2\theta-\sin^2\theta)}r\,\mathrm{d}\theta\,\mathrm{d}r$$
$$=2\iint_D xy\,\mathrm{e}^{x^2-y^2}\,\mathrm{d}x\,\mathrm{d}y$$
$$=2\int_0^1 x\,\mathrm{d}x\int_0^{\sqrt{1-x^2}}y\,\mathrm{e}^{x^2-y^2}\,\mathrm{d}y$$
$$=2\cdot\frac{-1}{2}\int_0^1 x\,\mathrm{d}x\int_0^{\sqrt{1-x^2}}\mathrm{e}^{x^2-y^2}\,\mathrm{d}(x^2-y^2)$$
$$=-\int_0^1 x\,\mathrm{e}^{x^2-y^2}\Big|_0^{\sqrt{1-x^2}}\mathrm{d}x=-\int_0^1 x(\mathrm{e}^{2x^2-1}-\mathrm{e}^{x^2})\mathrm{d}x$$
$$=-\frac{1}{4}\int_0^1\mathrm{e}^{2x^2-1}\mathrm{d}(2x^2-1)+\frac{1}{2}\int_0^1\mathrm{e}^{x^2}\mathrm{d}(x^2)$$
$$=-\frac{1}{4}\mathrm{e}^{2x^2-1}\Big|_0^1+\frac{1}{2}\mathrm{e}^{x^2}\Big|_0^1=\frac{1}{4}(\mathrm{e}+\mathrm{e}^{-1})-\frac{1}{2}.$$

解法2 在极坐标下，选择先对 θ 积分后对 r 积分.

$$I=\int_0^1\mathrm{d}r\int_0^{\frac{\pi}{2}}r^3\mathrm{e}^{r^2\cos2\theta}\sin2\theta\,\mathrm{d}\theta$$
$$=-\frac{1}{2}\int_0^1 r\,\mathrm{d}r\int_0^{\frac{\pi}{2}}\mathrm{d}(\mathrm{e}^{r^2\cos2\theta})=-\frac{1}{2}\int_0^1 r\left[\mathrm{e}^{r^2\cos2\theta}\right]_0^{\frac{\pi}{2}}\mathrm{d}r$$
$$=-\frac{1}{2}\int_0^1 r(\mathrm{e}^{-r^2}-\mathrm{e}^{r^2})\mathrm{d}r=-\frac{1}{4}\int_0^1(\mathrm{e}^{-r^2}-\mathrm{e}^{r^2})\mathrm{d}r^2$$
$$\overset{r^2=t}{=\!=}-\frac{1}{4}\int_0^1(\mathrm{e}^{-t}-\mathrm{e}^{t})\mathrm{d}t=-\frac{1}{4}\left(-\mathrm{e}^{-t}\Big|_0^1-\mathrm{e}^{t}\Big|_0^1\right)$$
$$=\frac{1}{4}(\mathrm{e}^{-1}+\mathrm{e})-\frac{1}{2}.$$

(13) 解 积分区域 D 如图 6-54 所示，采用极坐标. 先对 θ 积分后对 r 积分，则

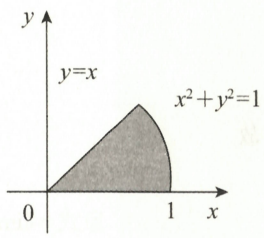

图 6-54

$$D=\left\{(r,\theta)\,\middle|\,0\leqslant r\leqslant1,0\leqslant\theta\leqslant\frac{\pi}{4}\right\}.$$
$$I=\int_0^1 r\,\mathrm{d}r\int_0^{\frac{\pi}{4}}\frac{r\cos\theta\cdot r\sin\theta}{1+r^2\cos^2\theta-r^2\sin^2\theta}\mathrm{d}\theta$$
$$=\frac{1}{2}\int_0^1 r\,\mathrm{d}r\int_0^{\frac{\pi}{4}}\frac{r^2\sin2\theta}{1+r^2\cos2\theta}\mathrm{d}\theta$$
$$=-\frac{1}{4}\int_0^1 r\,\mathrm{d}r\int_0^{\frac{\pi}{4}}\frac{\mathrm{d}(1+r^2\cos2\theta)}{1+r^2\cos2\theta}$$
$$=-\frac{1}{4}\int_0^1 r\,\mathrm{d}r\int_0^{\frac{\pi}{4}}\mathrm{d}[\ln(1+r^2\cos2\theta)]$$
$$=-\frac{1}{4}\int_0^1 r\ln(1+r^2\cos2\theta)\Big|_0^{\frac{\pi}{4}}\mathrm{d}r$$

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$$=-\frac{1}{4}\int_0^1 r[0-\ln(1+r^2)]\mathrm{d}r$$
$$=\frac{1}{8}\int_0^1 \ln(1+r^2)\mathrm{d}(1+r^2)$$
$$=\frac{1}{8}\left[(1+r^2)\ln(1+r^2)\Big|_0^1-\int_0^1\frac{1+r^2}{1+r^2}\cdot 2r\mathrm{d}r\right]$$
$$=\frac{1}{8}(2\ln 2-1).$$

【注】 若先对 r 积分后对 θ 积分,计算较烦琐.

(14) 解 $I=\int_0^1 \arcsin(2\sqrt{x-x^2})\mathrm{d}x\int_0^x \mathrm{d}y=\int_0^1 x\arcsin(2\sqrt{x-x^2})\mathrm{d}x$

$$=\int_0^1 x\arcsin\left[2\sqrt{\frac{1}{4}-\left(\frac{1}{2}-x\right)^2}\right]\mathrm{d}x$$
$$\xrightarrow{\frac{1}{2}-x=\frac{1}{2}\cos t}\frac{1}{4}\int_0^{\pi}(1-\cos t)\sin t\cdot\arcsin(\sin t)\mathrm{d}t$$
$$=\frac{1}{4}\int_0^{\frac{\pi}{2}} t(1-\cos t)\sin t\,\mathrm{d}t+\frac{1}{4}\int_{\frac{\pi}{2}}^{\pi}(\pi-t)(1-\cos t)\sin t\,\mathrm{d}t.$$

又 $\int_{\frac{\pi}{2}}^{\pi}(\pi-t)(1-\cos t)\sin t\,\mathrm{d}t\xrightarrow{\pi-t=u}\int_0^{\frac{\pi}{2}} u(1+\cos u)\sin u\,\mathrm{d}u$,故

$$I=\frac{1}{4}\int_0^{\frac{\pi}{2}} t(1-\cos t)\sin t\,\mathrm{d}t+\frac{1}{4}\int_0^{\frac{\pi}{2}} t(1+\cos t)\sin t\,\mathrm{d}t$$
$$=\frac{1}{2}\int_0^{\frac{\pi}{2}} t\sin t\,\mathrm{d}t=-\frac{1}{2}t\cos t\Big|_0^{\frac{\pi}{2}}+\frac{1}{2}\int_0^{\frac{\pi}{2}}\cos t\,\mathrm{d}t$$
$$=\frac{1}{2}\sin t\Big|_0^{\frac{\pi}{2}}=\frac{1}{2}.$$

【注】 在 $\left[\frac{\pi}{2},\pi\right]$ 上,$\arcsin(\sin t)=\pi-t$.

(15) 解
$$\int_{-\sqrt{t^2-x^2}}^{\sqrt{t^2-x^2}}\left[f(\sqrt{x^2+y^2})+2y\right]\mathrm{d}y$$
$$=2\int_0^{\sqrt{t^2-x^2}} f(\sqrt{x^2+y^2})\mathrm{d}y$$
$$\xrightarrow{\sqrt{x^2+y^2}=u}2\int_x^t\frac{uf(u)}{\sqrt{u^2-x^2}}\mathrm{d}u,$$

故

$$\text{原式}=\lim_{t\to 0^+}\frac{\int_0^t\mathrm{d}x\int_{-\sqrt{t^2-x^2}}^{\sqrt{t^2-x^2}}\left[f(\sqrt{x^2+y^2})+2y\right]\mathrm{d}y}{t^3}$$
$$=\lim_{t\to 0^+}\frac{2\int_0^t\mathrm{d}x\int_x^t\frac{uf(u)}{\sqrt{u^2-x^2}}\mathrm{d}u}{t^3}=2\lim_{t\to 0^+}\frac{\int_0^t uf(u)\mathrm{d}u\int_0^u\frac{1}{\sqrt{u^2-x^2}}\mathrm{d}x}{t^3}$$
$$=\pi\lim_{t\to 0^+}\frac{\int_0^t uf(u)\mathrm{d}u}{t^3}=\pi\lim_{t\to 0^+}\frac{tf(t)}{3t^2}=\frac{\pi}{3}.$$

(16) **解** 由已知，$f(0)=0$，$f(t)$ 是偶函数，只需讨论 $t>0$ 的情况. 用极坐标，有

$$f(t)=2\int_0^{2\pi}\mathrm{d}\theta\int_0^t r^3 f(r)\mathrm{d}r+t^4=4\pi\int_0^t r^3 f(r)\mathrm{d}r+t^4,$$

上式两边同时对 t 求导，得 $f'(t)=4\pi t^3 f(t)+4t^3$，且 $f(0)=0$，解此一阶线性微分方程，得

$$f(t)=\frac{1}{\pi}(\mathrm{e}^{\pi t^4}-1),t\geqslant 0.$$

而 $f(t)$ 是偶函数，故在 $(-\infty,+\infty)$ 内有 $f(t)=\frac{1}{\pi}(\mathrm{e}^{\pi t^4}-1)$.

(17) **解** 交换积分顺序，如图 6-55 所示，

$$\int_0^{x^2}\mathrm{d}t\int_{\sqrt{t}}^{x}f(t,u)\mathrm{d}u=-\int_0^{x}\mathrm{d}u\int_0^{u^2}f(t,u)\mathrm{d}t,$$

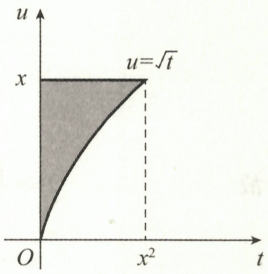

图 6-55

故

$$\text{原式}=\lim_{x\to 0^+}\frac{-\int_0^x\mathrm{d}u\int_0^{u^2}f(t,u)\mathrm{d}t}{\frac{1}{4}x^4}\overset{\text{洛必达}\atop\text{法则}}{=\!=\!=}\lim_{x\to 0^+}\frac{-\int_0^{x^2}f(t,x)\mathrm{d}t}{x^3}.$$

由积分中值定理，有

$$\int_0^{x^2}f(t,x)\mathrm{d}t=f(\xi,x)x^2(0\leqslant\xi\leqslant x^2),$$

故

$$\text{原式}=\lim_{x\to 0^+}\frac{-f(\xi,x)x^2}{x^3}.$$

由 $f(x,y)$ 在点 $(0,0)$ 处可微，则由可微的定义，有

$$f(\xi,x)=f(0,0)+f_x'(0,0)\xi+f_y'(0,0)x+o(\sqrt{x^2+\xi^2}),$$

又

$$\left|\frac{f_x'(0,0)\xi}{x}\right|\leqslant\left|\frac{f_x'(0,0)x^2}{x}\right|=|f_x'(0,0)x|,$$

则

$$\lim_{x\to 0^+}\frac{f_x'(0,0)\xi}{x}=0,\ \lim_{x\to 0^+}\frac{o(\sqrt{x^2+\xi^2})}{x}=\lim_{x\to 0^+}\frac{o(x)}{x}=0,$$

故

$$\begin{aligned}\lim_{x\to 0^+}\frac{-f(\xi,x)x^2}{x^3}&=\lim_{x\to 0^+}\frac{f(0,0)+f_x'(0,0)\xi+f_y'(0,0)x+o(\sqrt{x^2+\xi^2})}{-x}\\&=-f_y'(0,0)=-1.\end{aligned}$$

(18) **解** 设 $\iint_D f(u,v)\mathrm{d}u\mathrm{d}v=A$（$A$ 为一个数）. 在已知等式两边同时取二重积分，得

$$\begin{aligned}A&=\iint_D f(x,y)\mathrm{d}x\mathrm{d}y\\&=\iint_D(x^2+y^2-x+y-1)\mathrm{d}x\mathrm{d}y+A\iint_D\mathrm{d}x\mathrm{d}y.\end{aligned}$$

如图 6-56 所示，D 关于直线 $x=y$ 对称，则

$$\begin{aligned}A&=\frac{1}{2}\iint_D(x^2+y^2-x+y-1+y^2+x^2-y+x-1)\mathrm{d}x\mathrm{d}y+A\cdot\frac{\pi}{4}\cdot 2^2\\&=\iint_D(x^2+y^2-1)\mathrm{d}x\mathrm{d}y+\pi A\\&=\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^2(r^2-1)r\mathrm{d}r+\pi A=\pi+\pi A,\end{aligned}$$

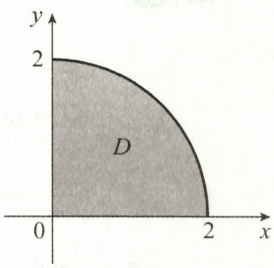

图 6-56

故 $A=\frac{\pi}{1-\pi}$，所求函数为

$$f(x,y)=x^2+y^2-x+y-1+\frac{\pi}{1-\pi}.$$

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(19) **解** 由已知,积分区域 D 如图 6-57 阴影部分所示,采用极坐标.

$$I=\int_0^{\frac{3}{4}\pi}\mathrm{d}\theta\int_1^2\frac{r}{r^2(1+\sin^2\theta)}\cdot r\mathrm{d}r=\int_0^{\frac{3\pi}{4}}\mathrm{d}\theta\int_1^2\frac{\mathrm{d}r}{1+\sin^2\theta}=\int_0^{\frac{3\pi}{4}}\frac{\mathrm{d}\theta}{1+\sin^2\theta},$$

而

$$\begin{aligned}\int\frac{\mathrm{d}\theta}{1+\sin^2\theta}&=\int\frac{\frac{1}{\cos^2\theta}}{\frac{1}{\cos^2\theta}+\tan^2\theta}\mathrm{d}\theta\\&=\int\frac{\mathrm{d}(\tan\theta)}{1+2\tan^2\theta}=\frac{1}{\sqrt{2}}\int\frac{\mathrm{d}(\sqrt{2}\tan\theta)}{1+(\sqrt{2}\tan\theta)^2}\\&=\frac{1}{\sqrt{2}}\arctan(\sqrt{2}\tan\theta)+C,\end{aligned}$$

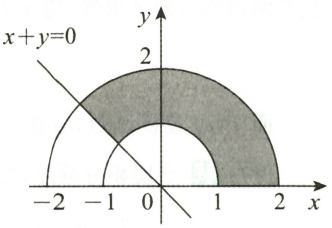

图 6-57

故

$$\begin{aligned}I&=\int_0^{\frac{3\pi}{4}}\frac{\mathrm{d}\theta}{1+\sin^2\theta}=\int_0^{\frac{\pi}{2}}\frac{\mathrm{d}\theta}{1+\sin^2\theta}+\int_{\frac{\pi}{2}}^{\frac{3\pi}{4}}\frac{\mathrm{d}\theta}{1+\sin^2\theta}\\&=\frac{1}{\sqrt{2}}\arctan(\sqrt{2}\tan\theta)\Big|_0^{\frac{\pi}{2}-0}+\frac{1}{\sqrt{2}}\arctan(\sqrt{2}\tan\theta)\Big|_{\frac{\pi}{2}+0}^{\frac{3\pi}{4}}\\&=\frac{1}{\sqrt{2}}\cdot\frac{\pi}{2}+\left(-\frac{1}{\sqrt{2}}\arctan\sqrt{2}\right)-\frac{1}{\sqrt{2}}\cdot\left(-\frac{\pi}{2}\right)\\&=\frac{\sqrt{2}}{2}\pi-\frac{\sqrt{2}}{2}\arctan\sqrt{2}.\end{aligned}$$

【注】 ①$\int_0^{\frac{3\pi}{4}}\frac{\mathrm{d}\theta}{1+\sin^2\theta}=\frac{1}{\sqrt{2}}\arctan(\sqrt{2}\tan\theta)\Big|_0^{\frac{3\pi}{4}}=-\frac{1}{\sqrt{2}}\arctan\sqrt{2}$ 是错误的. 原因是 $\frac{1}{\sqrt{2}}\arctan(\sqrt{2}\tan\theta)$ 在 $\theta=\frac{\pi}{2}\in\left[0,\frac{3}{4}\pi\right]$ 处无意义,它在 $\left[0,\frac{3\pi}{4}\right]$ 上不连续,即 $\frac{1}{\sqrt{2}}\arctan(\sqrt{2}\tan\theta)$ 不是被积函数在 $\left[0,\frac{3}{4}\pi\right]$ 上的原函数,不能直接用牛顿-莱布尼茨公式,要分段处理.

② 如果用换元法:令 $\tan\theta=u$,有

$$\int_0^{\frac{3\pi}{4}}\frac{\mathrm{d}\theta}{1+\sin^2\theta}=\int_0^{-1}\frac{\mathrm{d}u}{1+2u^2}=\frac{1}{\sqrt{2}}\arctan(\sqrt{2}u)\Big|_0^{-1}=-\frac{\sqrt{2}}{2}\arctan\sqrt{2}$$

也是错误的. 其原因是当 $t\in[-1,0]$ 时,$\theta=\arctan u$ 的值不落在原积分区间 $\left[0,\frac{3}{4}\pi\right]$ 上,不符合第二换元法条件要求.

(20) **解** 令 $A=\iint_D\frac{xf(x,y)}{x+y}\mathrm{d}x\mathrm{d}y$,则

$$\frac{xf(x,y)}{x+y}=\frac{x}{x+y}\sin\left(\pi\sqrt{x^2+y^2}\right)+\frac{A}{\pi}\cdot\frac{x}{x+y},$$

$$\begin{aligned}A&=\iint_D\frac{xf(x,y)}{x+y}\mathrm{d}x\mathrm{d}y=\iint_D\frac{x}{x+y}\sin\left(\pi\sqrt{x^2+y^2}\right)\mathrm{d}x\mathrm{d}y+\frac{A}{\pi}\iint_D\frac{x}{x+y}\mathrm{d}x\mathrm{d}y\\&=\iint_D\frac{y}{x+y}\sin\left(\pi\sqrt{x^2+y^2}\right)\mathrm{d}x\mathrm{d}y+\frac{A}{\pi}\iint_D\frac{y}{x+y}\mathrm{d}x\mathrm{d}y\\&=\frac{1}{2}\iint_D\sin\left(\pi\sqrt{x^2+y^2}\right)\mathrm{d}x\mathrm{d}y+\frac{A}{2\pi}\iint_D\mathrm{d}x\mathrm{d}y\\&=\frac{1}{2}\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_2^3\sin(\pi r)r\mathrm{d}r+\frac{A}{2\pi}\cdot\frac{5\pi}{4}=\frac{5}{4}+\frac{5A}{8},\end{aligned}$$

解得 $A=\dfrac{10}{3}$. 故

$$\iint_D f(x,y)\mathrm{d}x\mathrm{d}y=\iint_D \sin\left(\pi\sqrt{x^2+y^2}\right)\mathrm{d}x\mathrm{d}y+\frac{A}{\pi}\iint_D \mathrm{d}x\mathrm{d}y=\frac{5}{2}+\frac{10}{3\pi}\cdot\frac{5\pi}{4}=\frac{20}{3}.$$

(21) **解** 积分区域 D 如图 6-58 所示，设其密度为常数 ρ，考虑 D 的对称性，质心在 $x=\pi$ 上，即 $\overline{x}=\pi$，只需求

$$\overline{y}=\frac{\iint_D y\rho\mathrm{d}x\mathrm{d}y}{\iint_D \rho\mathrm{d}x\mathrm{d}y}=\frac{\iint_D y\mathrm{d}x\mathrm{d}y}{\iint_D \mathrm{d}x\mathrm{d}y}.$$

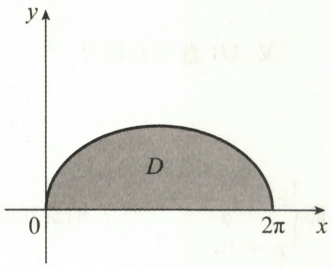

图 6-58

而面积

$$\iint_D \mathrm{d}x\mathrm{d}y=\int_0^{2\pi} y(x)\mathrm{d}x=\int_0^{2\pi}(1-\cos t)\cdot(1-\cos t)\mathrm{d}t=3\pi,$$

故

$$\overline{y}=\frac{1}{3\pi}\iint_D y\mathrm{d}x\mathrm{d}y=\frac{1}{3\pi}\int_0^{2\pi}\mathrm{d}x\int_0^{y(x)} y\mathrm{d}y=\frac{1}{6\pi}\int_0^{2\pi}[y(x)]^2\mathrm{d}x$$
$$=\frac{1}{6\pi}\int_0^{2\pi}(1-\cos t)^3\mathrm{d}t=\frac{5}{6},$$

所以质心为 $\left(\pi,\dfrac{5}{6}\right)$.

(22) **解** 考虑到被积函数 $3x^2+5y^2+7z^2$ 关于 z 是偶函数，对 V 补上下半球体得 $V_1: x^2+y^2+z^2\leqslant R^2$（见图 6-59），则 V_1 关于直线 $x=y=z$ 对称，故

$$\iiint_{V_1} x^2\mathrm{d}V=\iiint_{V_1} y^2\mathrm{d}V=\iiint_{V_1} z^2\mathrm{d}V=\frac{1}{3}\iiint_{V_1}(x^2+y^2+z^2)\mathrm{d}V,$$

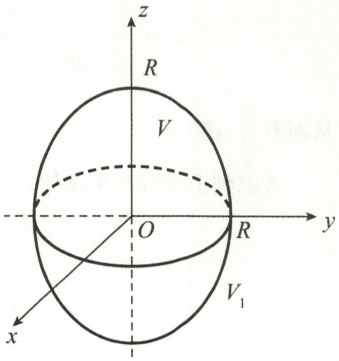

图 6-59

于是

$$I=\iiint_V(3x^2+5y^2+7z^2)\mathrm{d}V=\frac{1}{2}\iiint_{V_1}(3x^2+5y^2+7z^2)\mathrm{d}V$$
$$=\frac{1}{2}\left(3\iiint_{V_1} x^2\mathrm{d}V+5\iiint_{V_1} y^2\mathrm{d}V+7\iiint_{V_1} z^2\mathrm{d}V\right)$$
$$=\frac{15}{2}\iiint_{V_1} x^2\mathrm{d}V=\frac{15}{2}\cdot\frac{1}{3}\iiint_{V_1}(x^2+y^2+z^2)\mathrm{d}V$$
$$=\frac{5}{2}\iiint_{V_1}(x^2+y^2+z^2)\mathrm{d}V\overset{\text{球坐标}}{=\!=}\frac{5}{2}\int_0^{2\pi}\mathrm{d}\theta\int_0^{\pi}\mathrm{d}\varphi\int_0^{R} r^4\sin\varphi\mathrm{d}r$$
$$=\frac{5}{2}\cdot 2\pi\int_0^{\pi}\sin\varphi\mathrm{d}\varphi\int_0^{R} r^4\mathrm{d}r=2\pi R^5.$$

(23) **解** 去绝对值. $z=\sqrt{x^2+y^2}$ 将 V 分成两部分（见图 6-60）. 利用柱面坐标，得

$$V_1:\begin{cases}z^2\geqslant x^2+y^2,\\ z=0,\\ z=1,\end{cases}\qquad V_2:\begin{cases}z^2\leqslant x^2+y^2\leqslant 2,\\ z=0,\\ z=1,\end{cases}$$

即

$$V_1: 0\leqslant\theta\leqslant 2\pi,\ 0\leqslant r\leqslant 1,\ r\leqslant z\leqslant 1,$$
$$V_2: 0\leqslant\theta\leqslant 2\pi,\ 0\leqslant r\leqslant 1,\ 0\leqslant z\leqslant r$$

及

$$0\leqslant\theta\leqslant 2\pi,\ 1\leqslant r\leqslant\sqrt{2},\ 0\leqslant z\leqslant 1,$$

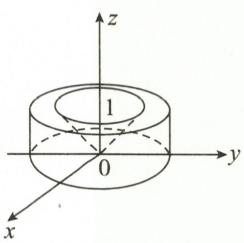

图 6-60

故

$$I=\iiint_{V_1}(z-\sqrt{x^2+y^2})\mathrm{d}x\mathrm{d}y\mathrm{d}z+\iiint_{V_2}(\sqrt{x^2+y^2}-z)\mathrm{d}x\mathrm{d}y\mathrm{d}z$$
$$=\int_0^{2\pi}\mathrm{d}\theta\int_0^1 r\mathrm{d}r\int_r^1(z-r)\mathrm{d}z+\int_0^{2\pi}\mathrm{d}\theta\int_0^1 r\mathrm{d}r\int_0^r(r-z)\mathrm{d}z+\int_0^{2\pi}\mathrm{d}\theta\int_1^{\sqrt{2}} r\mathrm{d}r\int_0^1(r-z)\mathrm{d}z$$

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$$=\frac{1}{6}(8\sqrt{2}-5)\pi.$$

(24) **解** 建立如图 6-61 所示的直角坐标系,将球心置于原点,球底面放在 xOy 上,则球面方程为

$$z=\sqrt{a^2-x^2-y^2}.$$

又 AB 直线方程为

$$\begin{cases}\dfrac{y}{a}+\dfrac{z}{-3}=1,\\ x=0,\end{cases}$$

即 $\begin{cases}z=\dfrac{3}{a}y-3,\\ x=0,\end{cases}$ 绕 z 轴旋转得到 $z=\dfrac{3}{a}\sqrt{x^2+y^2}-3$,即为圆锥面方程.

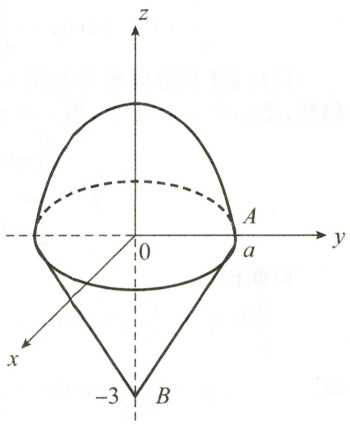

图 6-61

由已知条件,质心坐标 $(\overline{x},\overline{y},\overline{z})=(0,0,0)$,故

$$\overline{z}=\frac{\iiint\limits_V z\rho\,\mathrm{d}V}{\iiint\limits_V \rho\,\mathrm{d}V}=\frac{\iiint\limits_V z\,\mathrm{d}V}{\iiint\limits_V \mathrm{d}V}=0(\text{常数 }\rho\text{ 为物体的密度}),$$

从而有 $\iiint\limits_V z\,\mathrm{d}V=0$.

考虑到上部分为半球体,下部分为圆锥体,故分别采用球坐标和柱坐标计算,

$$\begin{aligned}\iiint\limits_V z\,\mathrm{d}V&=\int_0^{2\pi}\mathrm{d}\theta\int_0^{\frac{\pi}{2}}\mathrm{d}\varphi\int_0^a r\cos\varphi\cdot r^2\sin\varphi\,\mathrm{d}r+\int_0^{2\pi}\mathrm{d}\theta\int_0^a r\,\mathrm{d}r\int_{\frac{3}{a}r-3}^0 z\,\mathrm{d}z\\&=2\pi\cdot\frac{1}{2}\cdot\frac{a^4}{4}-\pi\left(\frac{9}{4a^2}\cdot a^4-6a^2+\frac{9}{2}a^2\right)\\&=\frac{\pi}{4}a^2(a^2-3)=0,\end{aligned}$$

解得 $a=\sqrt{3}$.

(25) **证** 积分区域如图 6-62(a) 所示,

$$\begin{aligned}\iint\limits_D f(x+y)\,\mathrm{d}x\,\mathrm{d}y&=\int_{-1}^0\mathrm{d}x\int_{-1-x}^{x+1}f(x+y)\,\mathrm{d}y+\int_0^1\mathrm{d}x\int_{x-1}^{1-x}f(x+y)\,\mathrm{d}y\\&\xlongequal{x+y=u}\int_{-1}^0\mathrm{d}x\int_{-1}^{1+2x}f(u)\,\mathrm{d}u+\int_0^1\mathrm{d}x\int_{2x-1}^1 f(u)\,\mathrm{d}u.\end{aligned}$$

如图 6-62(b) 所示,在 xOu 坐标下,交换积分顺序,则

$$\iint\limits_D f(x+y)\,\mathrm{d}x\,\mathrm{d}y=\int_{-1}^1\mathrm{d}u\int_{\frac{u-1}{2}}^{\frac{u+1}{2}}f(u)\,\mathrm{d}x=\int_{-1}^1 f(u)\,\mathrm{d}u.$$

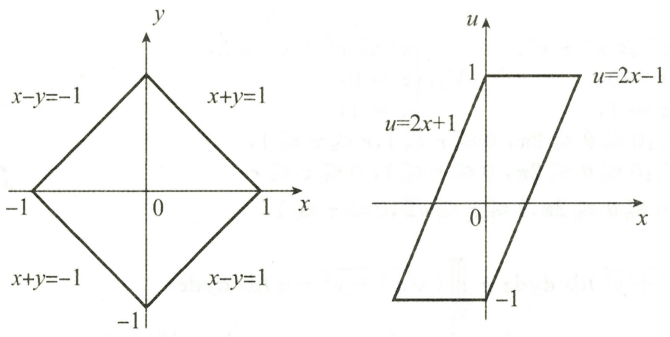

图 6-62

(26) 证 由已知，要证

$$\overline{x}=\frac{\iint\limits_D x\,\mathrm{d}x\mathrm{d}y}{\iint\limits_D \mathrm{d}x\mathrm{d}y}=\frac{\int_0^a \mathrm{d}x\int_0^{f(x)} x\,\mathrm{d}y}{\int_0^a \mathrm{d}x\int_0^{f(x)}\mathrm{d}y}=\frac{\int_0^a xf(x)\mathrm{d}x}{\int_0^a f(x)\mathrm{d}x}>\frac{3}{4}a,$$

即证$\int_0^a xf(x)\mathrm{d}x>\frac{3}{4}a\int_0^a f(x)\mathrm{d}x$.

令$F(a)=\int_0^a xf(x)\mathrm{d}x-\frac{3}{4}a\int_0^a f(x)\mathrm{d}x$，则

$$F'(a)=\frac{1}{4}af(a)-\frac{3}{4}\int_0^a f(x)\mathrm{d}x,\quad F''(a)=-\frac{1}{2}f(a)+\frac{1}{4}af'(a),$$

$$F'''(a)=-\frac{1}{4}f'(a)+\frac{1}{4}af''(a)=-\frac{1}{4}[f'(a)-f'(0)]+\frac{1}{4}af''(a)$$

$$=-\frac{1}{4}af''(\xi)+\frac{1}{4}af''(a)\quad(0<\xi<a)$$

$$=\frac{1}{4}a[f''(a)-f''(\xi)].$$

由$f''(x)$严格单调递增，知$F'''(a)>0$，故$F''(a)$单调递增.
又$F''(0)=0$，故$F''(a)>F''(0)=0$，$F'(a)$单调递增.
又$F'(0)=0$，故$F'(a)>F'(0)=0$，$F(a)$单调递增.
又$F(0)=0$，故$F(a)>F(0)=0$，所以不等式成立.

拓展题

解答题

(1) 证 (Ⅰ) 如图 6-63 所示，

$$\overline{x}=\frac{\iint\limits_D x\,\mathrm{d}x\mathrm{d}y}{\iint\limits_D \mathrm{d}x\mathrm{d}y}=\frac{\int_0^a x\,\mathrm{d}x\int_0^{f(x)}\mathrm{d}y}{\int_0^a \mathrm{d}x\int_0^{f(x)}\mathrm{d}y}$$

$$=\frac{\int_0^a xf(x)\mathrm{d}x}{\int_0^a f(x)\mathrm{d}x}=\frac{2}{3}a,$$

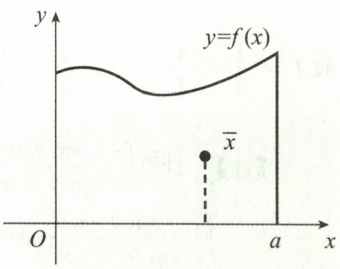

图 6-63

由$F(x)=\int_0^x f(t)\mathrm{d}t$，得$\frac{2}{3}x\cdot F(x)=\int_0^x tf(t)\mathrm{d}t$，等式两边同时对$x$求导，得

$$\frac{2}{3}x\cdot F'(x)+\frac{2}{3}F(x)=xf(x)=xF'(x),$$

即$F'(x)=\frac{2F(x)}{x}$.

解 (Ⅱ) 由(Ⅰ)知$F'(x)=\frac{2F(x)}{x}$，即

$$F'(x)-\frac{2}{x}F(x)=0,$$

解此一阶线性齐次微分方程，得通解为

$$F(x)=C_1\cdot e^{-\int\left(-\frac{2}{x}\right)\mathrm{d}x}=C_1x^2,$$

故$f(x)=F'(x)=2C_1x=Cx$(C为任意正常数).

(2) 解 D如图 6-64 所示，采用极坐标.

$x^2+y^2=\frac{1}{4}$的极坐标方程为$r=\frac{1}{2}$.

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$x^2+y^2=x^4+y^4$ 的极坐标方程为 $r=\sqrt{\dfrac{1}{\sin^4\theta+\cos^4\theta}}$.

$$\begin{aligned}I&=\iint_D\frac{xy}{x^2+y^2}\mathrm{d}x\,\mathrm{d}y\\&=\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_{\frac12}^{\sqrt{\frac{1}{\sin^4\theta+\cos^4\theta}}}\frac{r^2\sin\theta\cos\theta}{r^2}\cdot r\mathrm{d}r\\&=\int_0^{\frac{\pi}{2}}\sin\theta\cos\theta\cdot\frac12 r^2\Big|_{\frac12}^{\sqrt{\frac{1}{\sin^4\theta+\cos^4\theta}}}\mathrm{d}\theta\\&=\frac12\int_0^{\frac{\pi}{2}}\frac{\sin\theta\cos\theta}{\sin^4\theta+\cos^4\theta}\mathrm{d}\theta-\frac18\int_0^{\frac{\pi}{2}}\sin\theta\cos\theta\mathrm{d}\theta,\end{aligned}$$

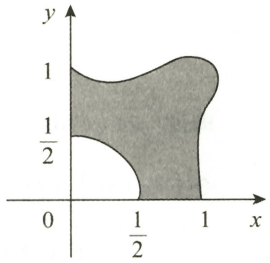

图 6-64

而
$$\int_0^{\frac{\pi}{2}}\sin\theta\cos\theta\mathrm{d}\theta=\int_0^{\frac{\pi}{2}}\sin\theta\mathrm{d}(\sin\theta)=\frac12\sin^2\theta\Big|_0^{\frac{\pi}{2}}=\frac12,$$

$$\begin{aligned}\int_0^{\frac{\pi}{2}}\frac{\sin\theta\cos\theta}{\sin^4\theta+\cos^4\theta}\mathrm{d}\theta&=\int_0^{\frac{\pi}{2}}\frac{2\sin\theta\cos\theta}{2(\sin^4\theta+\cos^4\theta)}\mathrm{d}\theta\\&=\int_0^{\frac{\pi}{2}}\frac{\sin 2\theta\mathrm{d}\theta}{2(\sin^4\theta+\cos^4\theta)+2\cos^2\theta\sin^2\theta-2\cos^2\theta\sin^2\theta}\\&=\int_0^{\frac{\pi}{2}}\frac{-\frac12\mathrm{d}(\cos 2\theta)}{(\sin^2\theta+\cos^2\theta)^2+(\cos^2\theta-\sin^2\theta)^2}\\&=-\frac12\int_0^{\frac{\pi}{2}}\frac{\mathrm{d}(\cos 2\theta)}{1+(\cos 2\theta)^2}=-\frac12\arctan(\cos 2\theta)\Big|_0^{\frac{\pi}{2}}\\&=-\frac12[\arctan(-1)-\arctan 1]=-\frac12\left(-\frac{\pi}{4}-\frac{\pi}{4}\right)=\frac{\pi}{4},\end{aligned}$$

故 $I=\dfrac12\times\dfrac{\pi}{4}-\dfrac18\times\dfrac12=\dfrac{1}{16}(2\pi-1)$.

【注】 计算 $\int_0^{\frac{\pi}{2}}\frac{\sin\theta\cos\theta}{\sin^4\theta+\cos^4\theta}\mathrm{d}\theta$ 也可以用以下方法:

$$\begin{aligned}\int_0^{\frac{\pi}{2}}\frac{\sin\theta\cos\theta}{\sin^4\theta+\cos^4\theta}\mathrm{d}\theta&\xlongequal{\sin\theta=t}\int_0^1\frac{t\,\mathrm{d}t}{t^4+(1-t^2)^2}=\frac12\int_0^1\frac{\mathrm{d}(t^2)}{2t^4-2t^2+1}\\&\xlongequal{t^2=u}\frac12\int_0^1\frac{\mathrm{d}u}{2u^2-2u+1}=\frac14\int_0^1\frac{\mathrm{d}u}{u^2-u+\frac12}\\&=\frac14\int_0^1\frac{\mathrm{d}\left(u-\frac12\right)}{\left(u-\frac12\right)^2+\left(\frac12\right)^2}=\frac14\times 2\arctan\frac{u-\frac12}{\frac12}\Big|_0^1\\&=\frac12[\arctan 1-\arctan(-1)]=\frac{\pi}{4}.\end{aligned}$$

(3) 证 (Ⅰ) 令 $F(x)=\int_0^x f(t)\mathrm{d}t$,则 $F'(x)=f(x)$,$F''(x)=f'(x)$,$F(0)=0$,$F(1)=0$.

对 $F(x)$ 应用泰勒公式,对于 $\forall a\in[0,1]$,有

$$F(x)=F(a)+F'(a)(x-a)+\frac{F''(\xi)}{2}(x-a)^2,\qquad①$$

其中 ξ 介于 a 与 x 之间. 将 $x=1$,$x=0$ 分别代入 ① 式,有

$$0=F(1)=F(a)+F'(a)(1-a)+\frac12F''(\xi_1)(1-a)^2,\qquad②$$

$$0=F(0)=F(a)+F'(a)(0-a)+\frac{1}{2}F''(\xi_2)(0-a)^2, \qquad ③$$

其中 ξ_1 介于 1 与 a 之间，ξ_2 介于 0 与 a 之间.

②－③，得

$$f(a)=F'(a)=-\frac{1}{2}\left[F''(\xi_1)(1-a)^2-F''(\xi_2)a^2\right].$$

故
$$|f(a)|=|F'(a)|\leqslant\frac{1}{2}\left[|F''(\xi_1)|(1-a)^2+|F''(\xi_2)|a^2\right]$$
$$=\frac{1}{2}\left[|f'(\xi_1)|(1-a)^2+|f'(\xi_2)|a^2\right].$$

令 $|f'(\xi)|=\max\{|f'(\xi_1)|,|f'(\xi_2)|\}$，$\xi\in(0,1)$，则

$$|f(a)|=|F'(a)|\leqslant\frac{1}{2}|f'(\xi)|\left[(1-a)^2+a^2\right]\leqslant\frac{1}{2}|f'(\xi)|\leqslant\frac{1}{2}M.$$

由 a 是 $[0,1]$ 上任一点，知 $|f(x)|\leqslant\frac{1}{2}M$，$x\in[0,1]$.

（Ⅱ）由于

$$\iint_D f(y)\mathrm{d}x\,\mathrm{d}y=\int_0^1\mathrm{d}y\int_y^1 f(y)\mathrm{d}x=\int_0^1(1-y)f(y)\mathrm{d}y$$
$$=\int_0^1\left(\frac{1}{2}-y\right)f(y)\mathrm{d}y+\frac{1}{2}\int_0^1 f(y)\mathrm{d}y$$
$$=\int_0^1\left(\frac{1}{2}-y\right)f(y)\mathrm{d}y,$$

故 $\left|\iint_D f(y)\mathrm{d}x\,\mathrm{d}y\right|\leqslant\frac{M}{2}\int_0^1\left|\frac{1}{2}-y\right|\mathrm{d}y\xlongequal{t=y-\frac{1}{2}}\frac{M}{2}\int_{-\frac{1}{2}}^{\frac{1}{2}}|t|\,\mathrm{d}t=M\int_0^{\frac{1}{2}}t\,\mathrm{d}t=\frac{M}{8}.$

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